

Power and Its Shadow: When Unanimity Serves the Hegemon

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Abstract

Why can unanimity benefit a powerful state when every member has the same veto? This paper studies a two-round bargaining game in which only weak states propose and a hegemon privately knows whether its disagreement payoff is low or high. Majority allows weak states to replace the hegemon's vote; unanimity makes its approval indispensable. Relative to a public-information benchmark, unanimity transfers informational rent to the low-outside-option type, while majority can eliminate that rent through exclusion. Some beliefs admit no equilibrium in the maintained pure-ballot class. An extension lets the hegemon propose first. Agenda power benefits it under unanimity, but majority can still be better when the outside option is low relative to the weak votes a winning coalition can exclude. The model offers a rationalist account of unequal bargaining power under formally equal rules.

1 Introduction

At the height of American power, the United States led the creation of the World Trade Organization (WTO). One would expect that bargaining outcomes in the trading system would reflect the asymmetry of power in the international system. Surely, International Organizations created by the hegemon would reflect such asymmetries in their rules. And yet, the WTO was created on the basis of equality rather than power: decisions by consensus meant that each member had equal veto power, and the US had no agenda power. Why would a hegemon help create an organization in which every state would formally be its equal?

The standard explanation is that equality under the law does not mean equality in practice. Coercion and Information institutions rule the negotiation process and shape the results, favoring the powerful over the weak (Steinberg 2002). The US and the (then) European Community drafted the negotiation texts used by all, offered market access and exit options to alter other states' alternatives, assembled linked packages, and gathered information through successive proposals and objections.

The best example is the process that ended with the creation of the WTO. At the close of the Uruguay Round, they threatened states that did not accept the creation of the WTO with withdrawal from the GATT 1947. The status quo was no longer available for such states. This episode shows how material power can enter a formally egalitarian institution through the agenda, the contract space, and the value of disagreement.

From the perspective of the distributive rationalist tradition, which focuses on bargaining theories of multilateral bodies (Voeten 2019), it is hard to reconcile this explanation with a rationalist framework. Consider the standard Baron and Ferejohn framework (Baron and Ferejohn 1989). Under open rule and equal recognition of the probability of proposal, combined with unanimity, there is no asymmetry of power and, as a result, no inequality of outcomes (Kalandrakis 2006). Concretely, if the US or the so-called Quad (US, EC, Japan, and Canada) had no formal agenda power, any country could propose a deal at any stage of a bargaining round. In fact, many coalitions did that during the Uruguay Round; for example, the agro-export group of countries called the Cairns Group (Cairns Group 1987). So, either a hegemon like the US made a blunder, or there is a need to further explain the institutional architecture of institutions like the OMC from a rationalist point of view, much like Fearon (1995) did for war.

More recently, the formal literature has investigated multilateral settings in which a single uninformed player is the sole proposer and multiple symmetric responders privately know their own types. On the one hand, they highlighted how the uninformed player may optimally choose to transfer informational rents as insurance against rejection (Glynia, Thum, and Xefteris 2026). On the other hand, they show that informed responders can likewise obtain informational rents by acting tough, since rejection under unanimity can improve their continuation offers (Piazolo and Vanberg 2025).

However, their models do not study settings in which power, rather than weakness, is concentrated in a single player, a key theme in International Relations. There is also no explanation of what happens to informational rent when the same actor holds both private information and proposal power, nor of the role of screening by uninformed weak players versus signaling by the informed player when proposing. Most importantly, in these models, changing the voting rule changes how many informed voters the proposer must buy and therefore the relative appeal of pooling and screening, but every winning coalition still includes an informed responder. They leave open a more fundamental question: what happens to the informed actor's bargaining power and informational rents when a minimum winning coalition can be formed entirely from uninformed voters? This question is central to understanding the comparison between majority and unanimity from the perspective of the most powerful actor.

The current paper fills this gap with a general n -player bargaining game in which power is concentrated in a single player, the hegemon, while all other players are symmetric and weak. As in the existing literature, the privately informed actor can be either a high or a low type. I first investigate what happens when even the powerful actor is stripped of proposal power. I then reintroduce proposal power by allowing the hegemon to make the first proposal, as often occurs in international organizations.

The paper's first contribution is to show that the voting rule determines whether the hegemon's private information has any bargaining value at all. Under majority, whenever the hegemon's vote is expensive relative to a weak state's, proposers assemble winning coalitions entirely from uninformed states and its informational rent is exactly zero. Under unanimity, the hegemon's approval is an essential input: proposals pool at the strong type's threshold, and the rent accrues to the hegemon

whose strength is overstated. Weak states cannot distinguish the two types, so they offer enough to satisfy the high outside option. The low type accepts the same package and keeps the difference as informational rent. The baseline also has a region of beliefs with no equilibrium in the maintained class of pure ballot strategies; no institutional payoff comparison is made in that region.

The second contribution reintroduces proposal power for the informed hegemon. Moving first gives the hegemon the familiar agenda rent under unanimity, but also changes the value of its private information. Majority benefits both types when the hegemon's outside option is low relative to the share of weak states that a winning coalition can leave out. In those cases, only the informational rent can reverse the public-information advantage of majority.

Section 3 works a numerical case. Section 4 defines the game and solution concept. Section 5 solves the complete-information benchmarks, then the private games, and finally the informational rents and their difference across voting rules. Section 6 then restores proposal power to the privately informed hegemon and decomposes the effect of the agenda stage into its complete-information component and its interaction with informational rent. Section 7 interprets the essential-input mechanism and its limits. The appendices give the proofs, the endpoint conventions, the exact correspondence tables, and the worked values.

2 Literature and contribution

2.1 Formal equality and informal power in international institutions

The paper begins from an established problem in international political economy. International organizations often give states formally equal rights while operating in a system with large disparities in markets, outside options, and coercive capacity (Krasner 1983; Koremenos, Lipson, and Snidal 2001). Research on informal governance shows how powerful states can preserve formally neutral rules while influencing agendas, staffing, enforcement, and the alternatives available to weaker members (Steinberg 2002; Stone 2011). The result is neither formal equality translated mechanically into equal influence nor material power operating wholly outside institutions. Power works through the bargaining environment in which formal rules are used.

Steinberg's study of the GATT and WTO supplies the closest substantive account (Steinberg 2002).

He distinguishes law-based bargaining, in which members take sovereign-equality procedures seriously, from power-based bargaining that invisibly weights outcomes toward the United States and the European Community. His historical evidence covers several practices: Green Room and Quad agenda setting, control of drafts and packages, market-access leverage, sanctions and forum exit, side payments and issue linkage, successive probes of state preferences, and legitimacy generated by an appearance of consent. These practices do not all constitute separate mechanisms. They can be organized by the strategic object each one changes.

Table 1: Steinberg’s practices and their strategic counterparts

Observed practice	Strategic object	Relation to this paper
Green Room, Quad, drafting, difficult-to-amend packages	Recognition, proposal, and amendment protocol	Familiar agenda power; deliberately removed from the baseline hegemon
Market size, sanctions, alternative forums, withdrawal from GATT 1947	Disagreement and continuation payoffs	Microfoundation for asymmetric and possibly privately known $o \in \{\ell, h\}$
Side payments, exceptions, transition periods, issue linkage, single undertaking	Feasible contract space and closure rule	Motivation for a divisible package; single undertaking is distinct from destruction of the old fallback
Proposals, objections, working groups, and ple-nary probes	Information structure	Steinberg emphasizes information flowing from weak members to agenda setters; the model studies private information held by the hegemon
Legitimacy, implementation, and reputational concerns	Future participation and enforcement	Complementary mechanisms outside the paper’s two-round baseline

This reconstruction clarifies the paper’s counterfactual. The model accepts the asymmetry of disagreement values and the availability of compensating concessions, both of which Steinberg helps motivate. It then removes the hegemon’s exclusive proposal power. That restriction does not assert that agenda power was absent from the WTO. It permits a cleaner question: can a formally equal

approval rule give a privately informed hegemon a distributive advantage even when weak states control proposals?

2.2 Proposal rights and multilateral bargaining

Baron and Ferejohn provide the canonical framework for that question (Baron and Ferejohn 1989). Their model places recognition, proposals, voting, and continuation values inside a sequential non-cooperative game. This move makes legislative outcomes depend on the rules governing both agenda formation and approval, rather than on voting power alone. It also establishes the vocabulary used here: recognized proposers divide a fixed surplus, build winning coalitions, and compare immediate agreement with continuation.

Kalandrakis isolates the distributive force of recognition more sharply (Kalandrakis 2006). For monotonic voting rules and arbitrary discount factors, appropriate proposal probabilities can support any division of the surplus as stationary equilibrium payoffs. Proposal rights can therefore dominate the political power attributed to votes or patience. This is the benchmark the present paper switches off. The hegemon is never recognized in the baseline, so a favorable payoff cannot be attributed to its probability of proposing. Work on proposal and veto rights and the broader legislative- bargaining literature further shows that approval thresholds, continuation values, and recognition jointly determine who is bought into a coalition (Winter 1996; McCarty 2000; Eraslan and Evdokimov 2019).

Heterogeneity in disagreement values supplies the complete-information bridge. Miller, Montero, and Vanberg show that a player with an expensive known vote can benefit under unanimity and be excluded from a minimal winning coalition under majority (Miller, Montero, and Vanberg 2018). That is a public-type effect of pivotality and coalition composition. The contribution here begins after this benchmark by asking what private information adds to the same institutional comparison.

2.3 Private information and the price of approval

Two recent models identify neighboring mechanisms that the present analysis does not claim as new. Piazzolo and Vanberg show how the voting rule changes the signaling value of rejection in a dynamic game (Piazzolo and Vanberg 2025). Glynia, Thum, and Xefteris show how it changes an

uninformed proposer’s insurance problem when responders privately know their acceptance costs (Glynia, Thum, and Xefteris 2026). In both settings, every winning coalition still contains an informed responder.

The present paper instead concentrates private information in one asymmetric actor. The weak states that can substitute for the hegemon are uninformed. Majority can therefore bypass the only private information relevant to the proposed payment, whereas unanimity makes that information unavoidable.

The distinction produces two results absent from the neighboring models. First, the approval rule acts as an on-off switch for the informed participant rather than only changing the price of one informed vote relative to another. Second, the public-type benchmark permits a type-specific decomposition between the value of a necessary vote and the additional rent created by uncertainty. In the pooling region, that rent can arise through immediate agreement, without an on-path rejection. The paper therefore studies who receives the surplus when the only privately informed actor moves from replaceable to indispensable.

Related models examine majority-only screening, cheap talk, reputational toughness, informational loss, and the aggregation of signals distributed across voters (Tsai 2009; Tsai and Yang 2010; Chen and Eraslan 2013, 2014; Ma 2023; Feddersen and Pesendorfer 1998). Here weak states hold no private signals about the hegemon. The institutional comparison operates through coalition composition and the price of one privately informed actor’s participation.

3 A working numerical illustration

Consider one hegemon and four weak states, with discount factor 0.9, low and high disagreement payoffs 0.10 and 0.35, and probability 0.80 of the high payoff. The terminal unanimity cutoff is

$$p^* = \frac{0.35 - 0.10}{1 - 0.10} = 0.2778.$$

The corresponding majority screening–exclusion cutoff is 0.2935. At the stated probability, private majority therefore excludes the hegemon, whereas private unanimity pools and gives both types 0.315.

The public payoff vectors under majority and unanimity are, respectively, $(0.09, 0.35)$ and $(0.09, 0.315)$. The private vectors are $(0.10, 0.35)$ and $(0.315, 0.315)$. Private information is consequently worth $(0.01, 0)$ under majority and $(0.225, 0)$ under unanimity. Consensus raises the low type's informational rent by 0.215 because weak states must pay the high threshold when they cannot replace the hegemon's vote.

For beliefs $0 < p \leq 0.2778$, the unanimity game has no equilibrium in the maintained pure-ballot class. The worked values illustrate one point at which both institutional correspondences exist; they are not an empirical calibration.

4 The model

4.1 Players, information, and proposals

There is one hegemon, H , and $m \geq 3$ weak states, $W = \{1, \dots, m\}$, for a total of $m + 1$ states. Nature selects H 's terminal disagreement payoff $o \in \{\ell, h\}$, observed only by H , with $\Pr(o = h) = p \in [0, 1]$. The two possible values satisfy

$$0 < \ell < h < 1.$$

The game has two rounds. In each round, one weak state is recognized uniformly to propose; H is never a proposer. The two recognition draws are independent and made with replacement. Every weak state remains eligible in Round 2, including the Round-1 proposer. A proposal by weak state i is the allocation vector

$$x = (x_H, (x_j)_{j \in W}) \in \mathcal{X}, \quad \mathcal{X} = \left\{ x : x_H \geq 0, x_j \geq 0, x_H + \sum_{j \in W} x_j \leq 1 \right\}.$$

Payments cannot be negative and side payments outside the proposed package are unavailable. Here x_H is the institutional concession assigned to H , and x_j is weak state j 's share, including the proposer when $j = i$. The fixed unit pie is exhausted on the equilibrium path. Agreement provides H no intrinsic benefit beyond x_H , which isolates informational concessions and pivotality from

any direct value of agreement.

4.2 Ballots, timing, and payoffs

The proposer is counted as voting yes. All other states vote simultaneously; the complete vote vector and outcome become public only after all votes have been cast. Define

$$k = \left\lfloor \frac{m+1}{2} \right\rfloor.$$

Under majority, passage requires k additional yes votes beyond the proposer's; unanimity requires every state's yes vote. The ballot protocol is otherwise identical across rules.

If a proposal passes, its allocations are implemented among the parties to the agreement. If H votes yes, it is a party and receives x_H . If H votes no but a majority passes the proposal without it, H is not a party: it receives its outside option o , and the share x_H is paid to no one, because the concession is specific to H and cannot be transferred to another player. Agreement and disagreement payoffs are therefore mutually exclusive at every history. In every equilibrium exclusion derived below the proposer chooses $x_H = 0$, and the hegemon receives exactly o .

If nothing passes in Round 1, no payoff is realized then and the game proceeds to Round 2. Round-2 payoffs are multiplied by $\beta \in (0, 1)$ when evaluated in Round-1 units. If nothing passes in Round 2, each weak state receives zero and H receives o in Round-2 units. A no vote is only a ballot action; it does not remove a player or trigger disagreement by itself. [AUTHOR: P1] The delayed terminal disagreement payoff represents the cost of prolonging international negotiations.

Table 2: Transitions and payoffs in the essential-input game

Event	Weak-state payoffs	H votes yes	H votes no
Proposal passes	Proposed x_j , including x_i	x_H	the outside option o alone; x_H is paid to no one, so the total distributed falls below one when $x_H > 0$
Round-1 proposal fails	No current payoff; proceed to Round 2	$\beta C_H(\mathfrak{h}^Y)$	$\beta C_H(\mathfrak{h}^N)$
Round-2 proposal fails	Zero	o	o

Here $\mathfrak{h}^Y$ and $\mathfrak{h}^N$ denote the distinct public histories in which H voted yes or no, and $C_H(\mathfrak{h}^a)$ denotes H 's Round-2 continuation value after history $\mathfrak{h}^a$. These continuation values need not be equal because the public vote can change beliefs before Round 2.

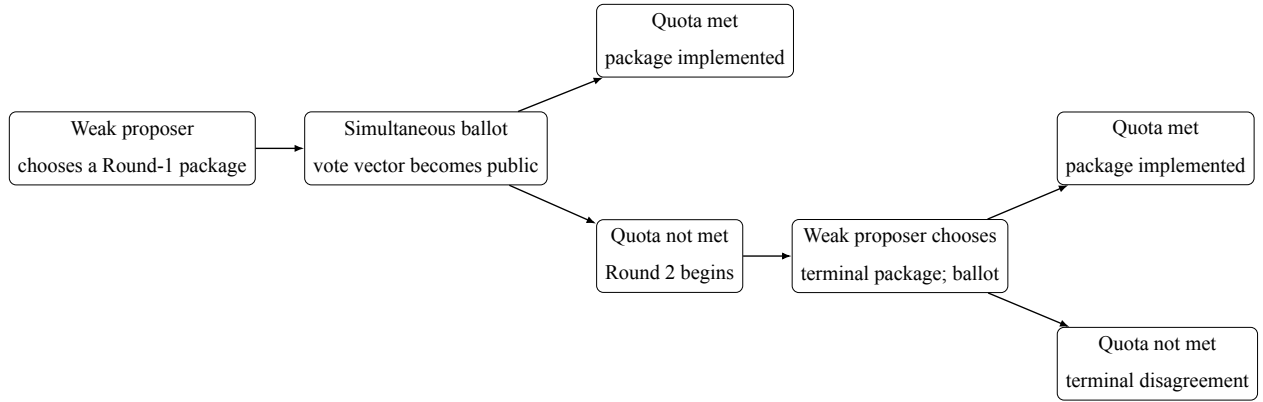


Figure 1: Sequence of proposals and ballots. Within each ballot, responding states vote simultaneously. A failed first-round proposal continues to the terminal round; a no vote alone does not end the game.

4.3 Solution concept

We use perfect Bayesian equilibrium with pure ballot strategies and three declared disciplines. First, an action by a player who does not know o does not change beliefs about o . After an action by

H , beliefs follow Bayes' rule whenever possible and otherwise satisfy structural consistency, the restriction stated exactly in Appendix A.2. Second, a weak responder evaluates yes and no as if its own vote were pivotal. Third, a voter accepts when exactly indifferent in expected value. Among proposal choices that give the proposer the same expected payoff, the selected proposal minimizes H 's expected payoff.

Endpoint beliefs preserve the support of the prior. If $p = 0$, the posterior assigns zero probability to h at every history; if $p = 1$, it assigns probability one to h . For $0 < p < 1$, an action by H whose Bayesian denominator is zero under the prescribed profile and the current belief may support any posterior in $[0, 1]$. This is a condition of our solution concept. It follows the no-signaling logic used in multistage games (Fudenberg and Tirole 1991); support preservation is analogous to, but narrower than, the never-dissuaded discipline discussed by Osborne and Rubinstein (1990). Tremble consistency motivates the restriction but is not invoked as a theorem for degenerate priors (Kreps and Wilson 1982).

Table 3: Scope of the formal results

Element	Maintained scope
Players and horizon	One hegemon, $m \geq 3$ weak states, two rounds
Agenda	Only weak states propose in both rounds
Voting	Simultaneous public ballots; majority or unanimity
Information	Private outside option $o \in \{\ell, h\}$; support-preserving end-points
Payoffs	Unit pie, $0 < \ell < h < 1$, $0 < \beta < 1$; no intrinsic agreement benefit for H
Equilibrium	Perfect Bayesian equilibrium in pure ballot strategies with the declared voting and proposal tie-break rules

4.4 Separate agenda-stage extension

We use superscript B for the two-round baseline and superscript A for the extension with the additional agenda stage. Thus the superscripts identify the arm of the comparison rather than a player

or an action.

The benchmark above remains the paper’s primary model. After solving it, we consider a separate extension in which the informed hegemon must make an earlier proposal at date A . Passage implements that proposal immediately; failure enters Round 1 of the benchmark and transports its complete continuation by exactly one factor β . The extension gives H no option to skip the proposal while retaining a date- A continuation. Section 6 states the economic results, and Appendices E–F preserve the complete equilibrium correspondences and date conventions.

5 Results

The solution proceeds by backward induction. We begin with public types because that benchmark separates the value of a necessary vote from the value of private information. We then solve the private terminal games, the private first-round games, and finally subtract the public benchmark component by component.

5.1 Complete-information benchmark

With a public type, no belief or signaling issue remains. The proposer compares the price of H ’s vote with the price of enough weak-state votes. In the terminal round, the relevant price of H is o . In the first round, it is βo , because rejection leads to the terminal round. Under majority, a weak-state vote costs β/m in first-round units. Under unanimity with public disagreement payoff o , each weak responder’s continuation price is $\beta(1 - o)/m$.

Proposition 5.1 (Public-type benchmark). *Fix a public type with disagreement payoff o . In the terminal round, majority passes a proposal without H , assigns $x_H = 0$, and gives the weak proposer the whole unit pie; H receives o . Terminal unanimity passes with $x_H = o$, and the weak proposer retains $1 - o$.*

In Round 1, unanimity passes immediately with $x_H = \beta o$. Majority includes H if $o \leq 1/m$ and excludes it if $o > 1/m$. At equality the proposal including H is selected. Consequently, the public

Round-1 payoff of H is

$$v_M^B(o) = \begin{cases} \beta o, & o \leq 1/m, \\ o, & o > 1/m, \end{cases} \quad v_U^B(o) = \beta o.$$

Table 4: Complete-information outcomes by round and voting rule

Round	Rule	Proposed payments	Weak proposer	H
2	Majority	$x_H = 0$; weak responders get zero	1	o
2	Unanimity	$x_H = o$; weak responders get zero	$1 - o$	o
1	Majority, $o \leq 1/m$	$x_H = \beta o$; $k - 1$ responders get β/m	$1 - \beta o - (k - 1)\beta/m$	βo
1	Majority, $o > 1/m$	$x_H = 0$; k responders get β/m	$1 - k\beta/m$	o
1	Unanimity	$x_H = \beta o$; every responder gets $\beta(1 - o)/m$	$1 - \beta o - (m - 1)\beta(1 - o)/m$	βo

At $o = 1/m$, buying H 's vote and buying the substitute weak-state vote cost the proposer the same amount. The proposal tie-break selects inclusion because it gives H the lower payoff $\beta o < o$.

5.2 Private information in the terminal round

Let

$$p^* = \frac{h - \ell}{1 - \ell}.$$

Under terminal majority, H 's vote is unnecessary and the belief never enters the proposer's choice. Under terminal unanimity, the proposer compares a low offer that succeeds only when the type is low with a high offer that both types accept.

Proposition 5.2 (Private terminal games). *Under majority, the unique equilibrium outcome assigns*

$x_H = 0$, pays no responding weak state, and gives the full pie to the proposer. The proposal passes without H ; a hegemon with outside option o receives o , and a weak state receives $1/m$ before recognition.

Under unanimity, if $p \leq p^*$, the proposer offers $x_H = \ell$. The low outside-option type accepts and the high outside-option type rejects, so the proposal succeeds with probability $1 - p$. If $p > p^*$, the proposer offers $x_H = h$ and both types accept. At $p = p^*$, the low offer is selected.

The cutoff equates $(1 - p)(1 - \ell)$, the proposer's payoff from the low offer, with $1 - h$, its payoff from pooling. Notice that no discount factor appears inside this terminal comparison.

5.3 Private majority in Round 1

Let $w = \beta/m$. The proposer's payoffs from the three undominated outcome classes are:

$$\begin{aligned}\Pi_E &= 1 - kw, \\ \Pi_S(p) &= (1 - p) [1 - (k - 1)w - \beta\ell] + pw, \\ \Pi_P &= 1 - (k - 1)w - \beta h.\end{aligned}$$

Exclusion buys k weak responders at price w each and sets $x_H = 0$. Screening buys $k - 1$ weak responders and offers $\beta\ell$: the low outside-option type accepts and the high outside-option type delays. Pooling buys $k - 1$ weak responders and offers βh , which both types accept. Deliberate delay is strictly worse than exclusion because $1 - \beta(k + 1)/m > 0$.

When $\ell < 1/m$, define

$$p_{S=E} = \frac{\beta(1/m - \ell)}{\beta(1/m - \ell) + 1 - \beta(k + 1)/m}.$$

When $h < 1/m$, define

$$p_{S=P} = \frac{\beta(h - \ell)}{1 - \beta\ell - \beta k/m}.$$

Proposition 5.3 (Private majority correspondence). *For $m \geq 3$, the equilibrium outcome class is:*

1. If $h < 1/m$, screening for $p \leq p_{S=P}$ and pooling above it.

2. If $\ell < 1/m < h$, screening for $p \leq p_{S=E}$ and exclusion above it.
3. If $1/m < \ell < h$, exclusion for every p .
4. If $\ell = 1/m < h$, screening at $p = 0$ and exclusion for $p > 0$.
5. If $\ell < h = 1/m$, screening for $p \leq p_{S=E}$. Above that cutoff, exclusion is selected when $(1 - p)\ell + ph < \beta h$, pooling is selected when the inequality is reversed, and the entire proposal segment joining the two survives when the two expressions are equal.

At every cutoff involving screening, screening is selected. Permuting the identities of identically paid weak responders generates additional equilibria but does not change H 's payoff vector or the outcome class. Payoffs of weak states identified by label can be permuted and, on the residual proposal family, vary with the common proposal weight.

The last segment is genuine multiplicity. Its mixing weight is a probability over pure proposals in the proposer's strategy, and the same weight must be used jointly for payoffs and outcomes. It is not a license to combine componentwise minima and maxima into an unattained rectangle.

5.4 Private unanimity in Round 1

Let

$$w_\ell^U = \frac{\beta(1 - \ell)}{m}, \quad w_h^U = \frac{\beta(1 - h)}{m}.$$

At $p = 0$, support preservation prevents an impossible high type from being reintroduced by an off-path belief. The proposer can therefore offer the low threshold $\beta\ell$, pay each responding weak state its continuation w_ℓ^U , and keep $w_\ell^U + 1 - \beta$. When $p > p^*$, the continuation itself pools, so the proposer offers βh , pays each responder w_h^U , and keeps $w_h^U + 1 - \beta$.

The intermediate cell fails for a different reason. A feasible deviation can pay every weak responder enough to force a yes vote. Once all weak votes are yes, no pure contingent voting profile for the two possible types of H survives: pooling yes is rejected by the high type, pooling no is overturned by the low type at equality, and either separating profile gives one type a profitable imitation.

Proposition 5.4 (Private unanimity correspondence). *Under unanimity:*

- $p = 0$: immediate low-type agreement with $x_H = \beta\ell$, $x_j = w_\ell^U$, $x_i = w_\ell^U + 1 - \beta$;
 $0 < p \leq p^*$: no perfect Bayesian equilibrium in pure ballot strategies;
 $p^* < p \leq 1$: immediate pooling agreement with $x_H = \beta h$, $x_j = w_h^U$, $x_i = w_h^U + 1 - \beta$.

The corresponding payoff vectors for H , ordered as low and high type, are $(\beta\ell, \beta h)$ at $p = 0$ and $(\beta h, \beta h)$ above p^* .

Remark (Pure-strategy scope). The middle cell establishes nonexistence only within the paper's maintained space of pure ballot strategies. We do not derive, characterize, select, or compare equilibria involving mixed ballot strategies, and make no claim about existence under such strategies.

Table 5: Private-information Round-1 equilibrium correspondences

Rule	Parameter cell	Selected outcome class	H payoff vector
Majority	$h < 1/m$	Screening for $p \leq p_{S=P}$; pooling above	$(\beta\ell, \beta h)$ or $(\beta h, \beta h)$
Majority	$\ell < 1/m < h$	Screening for $p \leq p_{S=E}$; exclusion above	$(\beta\ell, \beta h)$ or (ℓ, h)
Majority	$1/m < \ell$	Exclusion for every p	(ℓ, h)
Majority	$\ell = 1/m < h$	Screening at $p = 0$; exclusion for $p > 0$	$(\beta\ell, \beta h)$ or (ℓ, h)
Majority	$\ell < h = 1/m$	Screening through $p_{S=E}$; above, exclusion, pooling, or their exact proposal segment under the tie-break	$(\beta\ell, \beta h)$, (ℓ, h) , $(\beta h, \beta h)$, or the linked segment
Unanimity	$p = 0$	Immediate low-type agreement	$(\beta\ell, \beta h)$
Unanimity	$0 < p \leq p^*$	No PBE in pure ballot strategies	$\emptyset$
Unanimity	$p^* < p \leq 1$	Immediate pooling agreement	$(\beta h, \beta h)$

Prices and coalition anatomy

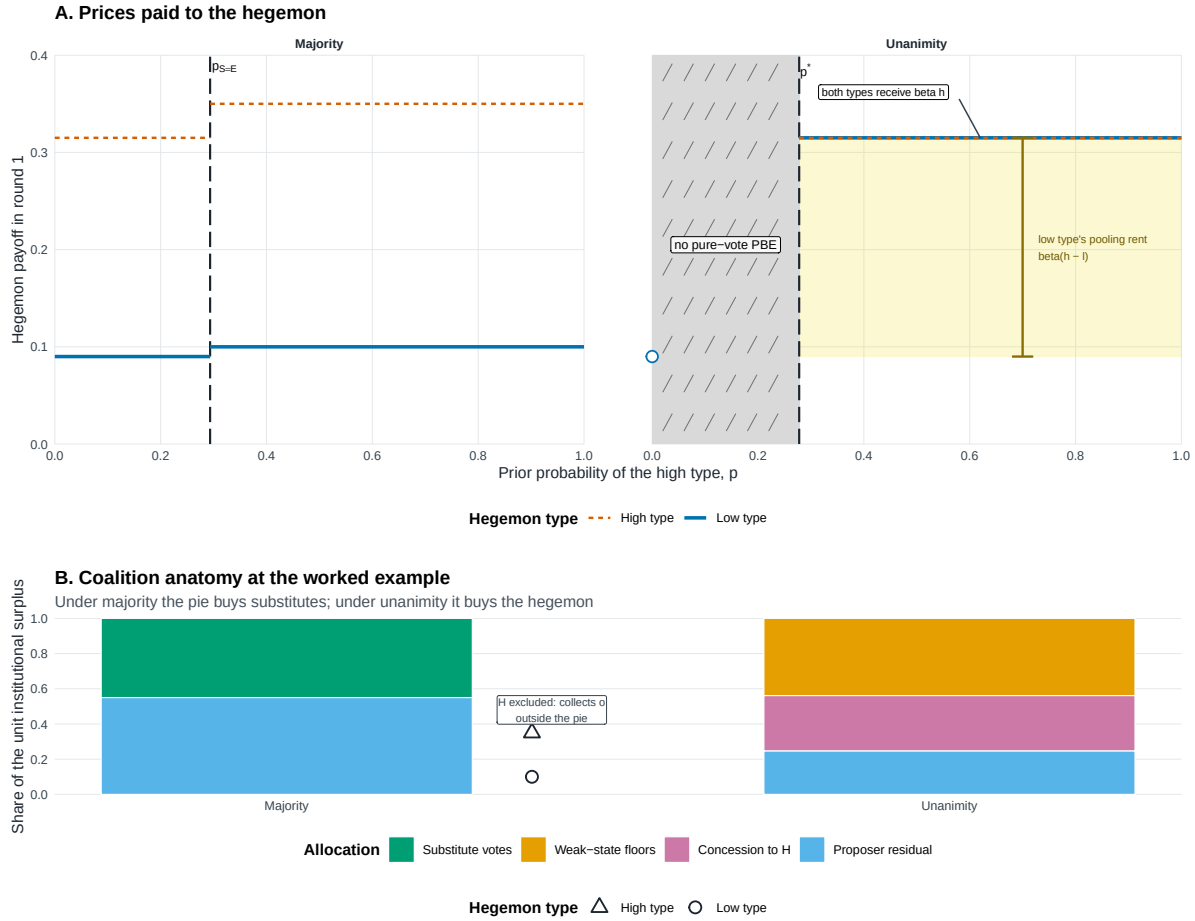


Figure 2: The price and coalition anatomy of the private games. Majority can replace the hegemon's vote with an additional weak-state vote; unanimity cannot. Labels report first-round prices and preserve the distinct low-type, pooling, and exclusion outcomes.

5.5 Comparing the private games

The comparison below uses the same parameter point under both institutions and only cells in which both games have a pure-strategy equilibrium. The timing wedges are $(1 - \beta)\ell$ and $(1 - \beta)h$: exclusion realizes the disagreement payoff now, whereas inclusion can price a continuation that is discounted in Round-1 units. They are distinct from the discounted type gap $\beta(h - \ell)$ and the

cross-date quantity $\beta h - \ell$.

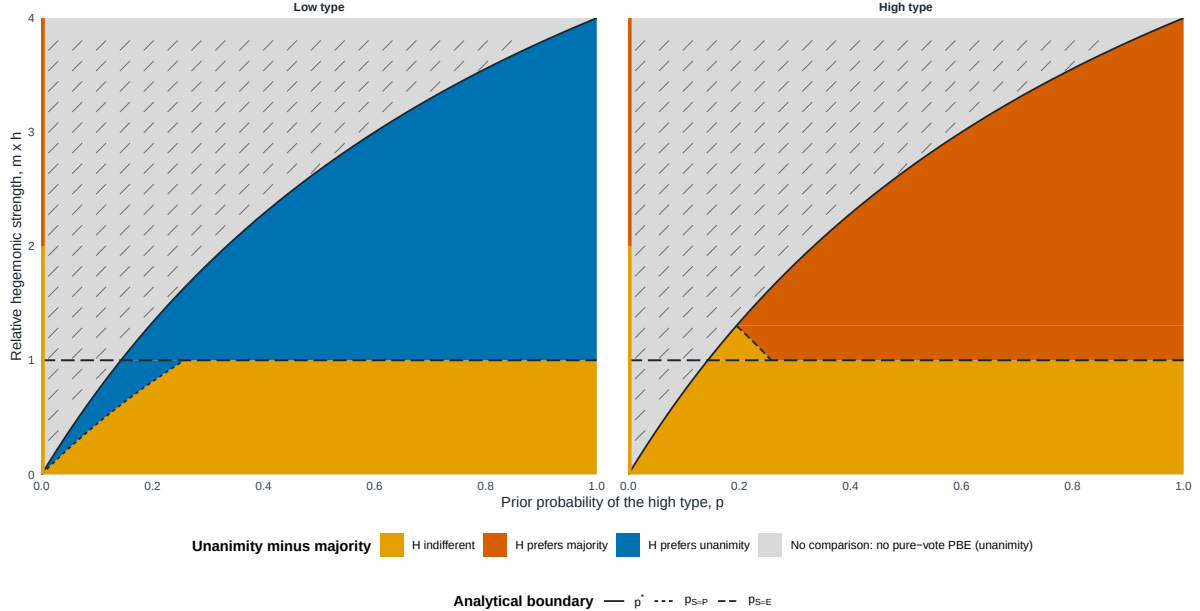
Proposition 5.5 (Private institutional payoff contrast). *Write $V_U^B - V_M^B$ as a low-type/high-type vector.*

1. *At $p = 0$, the vector is $(0, 0)$ if majority screens and $-(1 - \beta)\ell, -(1 - \beta)h$ if majority excludes.*
2. *For $0 < p \leq p^*$, the contrast is empty because private unanimity has no pure-strategy equilibrium.*
3. *For $p > p^*$, the vector is $(\beta(h - \ell), 0)$ if majority screens, $(0, 0)$ if majority pools, and $(\beta h - \ell, -(1 - \beta)h)$ if majority excludes.*
4. *On the residual exclusion–pooling segment at $h = 1/m$, the exact attainable set is $\{\lambda(\beta h - \ell, -(1 - \beta)h) : \lambda \in [0, 1]\}$, restricted to the proposal weights that survive the tie-break. The same λ binds payoffs and outcomes.*

Thus unanimity does not have a single unconditional payoff effect. Its effect depends on which substitute coalition disciplines the majority proposer and on which type is evaluated. Figure 3 maps these exact classes without averaging types.

The private-rule payoff contrast is type-specific

Closed-form slice $l = 0.50 \times h$; $m = 4$ weak states; $\beta = 0.90$



Colored regions compare the hegemon's private-information payoff under unanimity and majority at the same parameter point. The hatched region is empty because unanimity has no perfect Bayesian equilibrium in pure ballot strategies. The left-edge segment is the literal $p = 0$ endpoint. The horizontal threshold $m \times h = 1$ is the cost at which majority switches between buying the hegemon and buying an additional weak-state vote. No prior-weighted or ex ante image is displayed. Note: model-generated closed-form regions.

Figure 3: Private-game payoff contrast, unanimity minus majority, by hegemon type. Each cell is evaluated at the same parameter point. The hatched band marks parameter values for which private unanimity has no pure-strategy equilibrium and the contrast is therefore empty.

5.6 Informational rents and the difference of differences

For rule $g \in \{M, U\}$, define the type-contingent informational-rent correspondence

$$IR_g^B = V_g^B - v_g^B,$$

where the public benchmark is evaluated at the same type and the same remaining parameters. The institutional informational-rent contrast is

$$\Delta IR^B = IR_U^B - IR_M^B.$$

These are differences between correspondences. If a source correspondence is empty, so is the difference. Under multiplicity, subtraction preserves the entire attainable set and its common proposal weight.

We write the three public outside-option regions in words: both types are included when $h \leq 1/m$; the low type is included and the high type excluded when $\ell \leq 1/m < h$; and both types are excluded when $1/m < \ell$.

The rent comparison first asks what private information changes *within* each rule. Under unanimity, the public benchmark already buys the known type at its discounted threshold; private pooling therefore raises only the low type's payoff. Under majority, information matters only when the private equilibrium changes the public inclusion decision or the price paid to a type.

Proposition 5.6 (Informational rents by voting rule). *Under unanimity,*

$$IR_U^B = \begin{cases} \{(0, 0)\}, & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ \{(\beta(h - \ell), 0)\}, & p > p^*. \end{cases}$$

Under majority, the informational rent depends jointly on the public region and the private equilibrium class:

<i>both types included, screening</i>	: $(0, 0)$,
<i>both types included, pooling</i>	: $(\beta(h - \ell), 0)$,
<i>both types included, exclusion</i>	: $((1 - \beta)\ell, (1 - \beta)h)$,
<i>low included, high excluded, screening</i>	: $(0, -(1 - \beta)h)$,
<i>low included, high excluded, exclusion</i>	: $((1 - \beta)\ell, 0)$,
<i>both types excluded, exclusion</i>	: $(0, 0)$.

On the residual exclusion–pooling segment at $h = 1/m$, the exact set is $\{\lambda((1 - \beta)\ell, (1 - \beta)h) + (1 - \lambda)(\beta(h - \ell), 0) : \lambda \in [0, 1]\}$, restricted to the segment that survives the proposal tie-break.

The difference of differences then removes majority's informational rent from unanimity's. Screening can leave a positive low-type contrast in the high-belief cell because unanimity pools. Exclusion instead compares the high threshold discounted to the present with the low current disagreement payoff, so its low-type sign is governed by $\beta h - \ell$.

Proposition 5.7 (Institutional informational-rent contrast). *At $p = 0$, $\Delta IR^B = (0, 0)$ when both*

types are included or both are excluded, while $\Delta IR^B = (0, (1-\beta)h)$ when the low type is included and the high type excluded; the high-type component there is off support but remains part of the type-contingent vector.

For $0 < p \leq p^*$, $\Delta IR^B = \emptyset$.

For $p > p^*$, the exact vectors are

both types included, screening	: $(\beta(h - \ell), 0)$,
both types included, pooling	: $(0, 0)$,
both types included, exclusion	: $(\beta h - \ell, -(1 - \beta)h)$,
low included, high excluded, screening	: $(\beta(h - \ell), (1 - \beta)h)$,
low included, high excluded, exclusion	: $(\beta h - \ell, 0)$,
both types excluded, exclusion	: $(\beta(h - \ell), 0)$.

On the residual segment, the exact attainable set is $\{\lambda(\beta h - \ell, -(1 - \beta)h) : \lambda \in [0, 1]\}$ over the surviving proposal weights.

In every residual-segment case, $\lambda \in [0, 1]$ is restricted to the proposal weights that survive the tie-break and binds all payoff and outcome components.

The sign pattern is transparent once each component is kept separate. Above the unanimity cutoff, the low type gains $\beta(h - \ell) > 0$ when majority screens because unanimity pools and pays the high threshold. The same screening comparison is zero at $p = 0$, while the middle cell is empty. When majority excludes, the low type's gain is $\beta h - \ell$, which is positive, zero, or negative according as βh is above, equal to, or below ℓ . The high type gains $(1 - \beta)h > 0$ only when private majority screens even though public majority would exclude it; it loses $(1 - \beta)h$ when public majority includes it but private majority excludes. When both private rules pool, the informational-rent contrast is exactly zero.

Power versus information

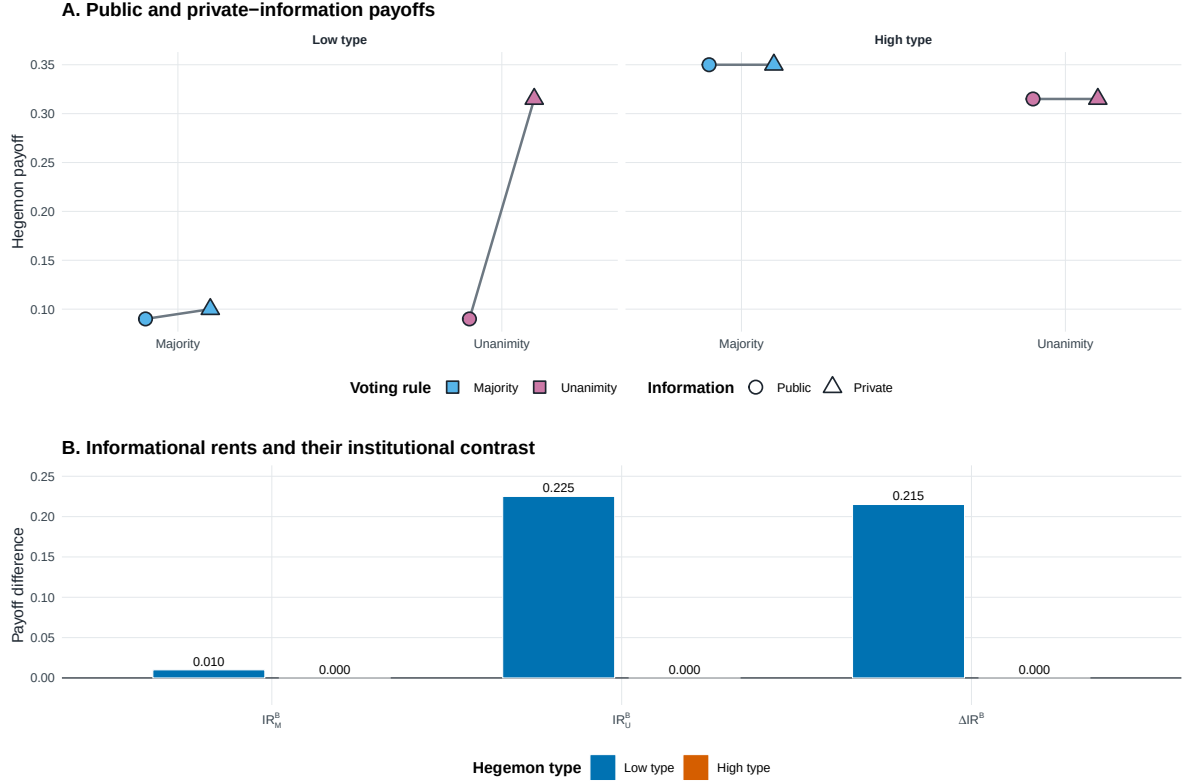


Figure 4: Power and information in the hegemon's payoff. The public benchmark captures the effect of making the hegemon's vote necessary when its type is known. Informational rent compares the private game with that benchmark, and ΔIR^B compares this rent across unanimity and majority without recombining types or selecting among multiple equilibria.

6 Agenda power and its informational shadow

The baseline removes an exclusive formal proposal privilege from H to identify informational power through pivotality. We now restore agenda power in a separate extension. Before the two-round game, at date A , the informed hegemon must propose a feasible division of the unit pie. If that proposal fails, the game enters Round 1 of the baseline. The continuation is therefore transported from Round 1 to date A by exactly one factor β . The proposal is mandatory: H cannot skip

the agenda stage and obtain the continuation at date A .

Let

$$e = m - k, \quad v_M^{\text{safe}} = 1 - \frac{k\beta}{m}.$$

Under majority, k is the number of weak votes that H must buy and e is the number of weak states that a minimal winning coalition can leave outside. The private games also retain the same off-path specification, summarized by the likelihood-ratio coordinate ρ and its implied posterior μ^{off} , where

$$\mu^{\text{off}} = \frac{p\rho}{1 - p + p\rho}$$

for an interior prior. At an endpoint, ρ is irrelevant and $\mu^{\text{off}} = p$. Comparisons across rules always pair complete equilibrium records from the same economy and under the same off-path specification. They never combine type payoffs, proposals, or outcomes from different records.

6.1 Public agenda power

Suppose first that o is public before H proposes. Under unanimity, H buys every weak vote at its continuation price and agreement is immediate. Under majority, it buys a minimal coalition when passage is better than continuation, but a type with a sufficiently high outside option may deliberately induce rejection.

Proposition 6.1 (Public agenda payoffs and institutional gap). *The resulting payoffs at date A are*

$$v_U^A(o) = 1 - \beta + \beta^2 o$$

and

$$v_M^A(o) = \begin{cases} 1 - \frac{k\beta(1 - \beta o)}{m}, & o \leq 1/m, \\ \max\{v_M^{\text{safe}}, \beta o\}, & o > 1/m. \end{cases}$$

In the first majority branch, any lottery over minimal coalitions is retained. For $o > 1/m$, define

$$o_M^* = \frac{v_M^{\text{safe}}}{\beta} = \frac{1}{\beta} - \frac{k}{m} > \frac{1}{m}.$$

Agreement is immediate for $o < o_M^$, any linked mixture of passage and delay is possible at equality, and every equilibrium delays for $o > o_M^*$. If $o_M^* \geq 1$, the last branch is outside the maintained domain $o \in (0, 1)$. Unanimity does not inherit this delay: buying all votes beats rejection by exactly $1 - \beta$.*

Define the public institutional gap with the common orientation unanimity minus majority,

$$\Delta v^A(o) = v_U^A(o) - v_M^A(o).$$

Then

$$\Delta v^A(o) = \begin{cases} -\beta \frac{e}{m}(1 - \beta o), & o \leq 1/m, \\ \beta \left(\beta o - \frac{e}{m} \right), & o > 1/m \text{ and } \beta o \leq v_M^{\text{safe}}, \\ (1 - \beta)(1 - \beta o), & o > 1/m \text{ and } \beta o \geq v_M^{\text{safe}}. \end{cases}$$

The last two expressions coincide at $\beta o = v_M^{\text{safe}}$. The first branch is negative; for $o > 1/m$, the sign of $\Delta v^A(o)$ is the sign of $\beta o - e/m$. Agenda power under public information thus need not favor unanimity. If $\beta h < e/m$, majority gives both types a strictly larger payoff; equality yields a weak ranking. Above that boundary, the public component can itself point toward unanimity.

Appendices E.6–E.8 derive the public equilibrium classes and the sign characterization. The result shows that proposal power alone does not produce a uniform institutional ranking.

Panel A: public institutional gap

$$\Delta v^A(o) = v_U^A(o) - v_M^A(o)$$

$$o \leq 1/m : \Delta v^A(o) < 0,$$

$$o > 1/m : \text{sgn } \Delta v^A(o) = \text{sgn}(\beta o - e/m).$$

Public information favors majority below the boundary and can favor unanimity above it.

Panel B: an informational reversal

Low type	M	U	$U - M$
Public	.5905	.1810	-.4095
Private	.5905	.7290	+.1385
Information	.0000	.5480	+.5480

Figure 5: Public power and an informational reversal. Panel A reports the exact sign boundary for $\Delta v^A(o)$. Panel B uses one linked member with $m = 4$, $\beta = .90$, $\ell = .10$, $h = .90$, and $p = .95$; the higher value $h = .90$, rather than the running example's $h = .35$, is used because an informational reversal requires $\beta h > e/m$. The unanimity value is the lower endpoint of its admissible pooling interval. The numbers are a theoretical illustration, not an empirical calibration or a global ranking of the correspondence.

6.2 Private agenda power and institutional comparison

Let $V_g^A = (V_g^A(\ell), V_g^A(h))$ denote the linked type-payoff vector generated by one complete private equilibrium record under rule $g \in \{M, U\}$. The majority correspondence contains pure pooling, separation, agreement, and deliberate-delay classes as well as Borel mixtures. It is not replaced by a selected equilibrium or by componentwise envelopes. Under unanimity, the exact payoff fibers can be written directly with the public value function $v_U^A(o) = 1 - \beta + \beta^2 o$:

$$V_U^A = \begin{cases} (v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\}), & p = 0, \\ (v_U^A(\ell), v_U^A(\ell)), & 0 < p \leq p^*, v_U^A(\ell) \geq \beta^2 h, \mu^{\text{off}} = 0, \\ \{(u, u) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1, \mu^{\text{off}} = 0, \\ (v_U^A(h), v_U^A(h)), & p^* < p < 1, \mu^{\text{off}} \in (p^*, 1], \\ (v_U^A(h), v_U^A(h)), & p = 1, \\ \emptyset, & \text{otherwise.} \end{cases}$$

Here and below $\emptyset$, also reported as none in the source correspondences, means an empty correspondence under the maintained pure-PBE M/S/B architecture. It is not a zero effect or a claim about mixed equilibria outside that architecture.

Institutional comparison begins with the exact fiber product

$$\mathcal{J}_A^{\text{bind}}(\mathbf{d}, \rho, \mu^{\text{off}}) = \mathcal{B}_M(\mathbf{d}, \rho, \mu^{\text{off}}) \times_{(\mathbf{d}, \rho, \mu^{\text{off}})} \mathcal{B}_U(\mathbf{d}, \rho, \mu^{\text{off}}).$$

For each member, define the private institutional contrast using the same unanimity-minus-majority orientation as the public contrast:

$$\Delta V^A(o) = V_U^A(o) - V_M^A(o), \quad \Delta V_{\text{ea}}^A = (1 - p)\Delta V^A(\ell) + p\Delta V^A(h).$$

Here and below, the subscript ea denotes the ex ante coordinate. The affine average is applied only after preserving the linked type vector. If either source is empty in the common fiber, the comparison is empty.

The exact correspondence may remain set-valued, but one selection-free region is available.

Proposition 6.2 (Selection-free majority advantage). *Fix a nonempty common fiber. If*

$$\beta h < \frac{e}{m},$$

then, for both types and ex ante,

$$V_U^A(o) - V_M^A(o) \leq -\beta \left(\frac{e}{m} - \beta h \right) < 0.$$

At equality the same ranking is weak.

This condition is sufficient, not necessary. It follows from the lower bound $V_M^A(o) \geq v_M^{\text{safe}}$ and the upper bound $V_U^A(o) \leq v_U^A(h)$; outside it, the full correspondence determines the ranking. Appendices B.7–B.8 prove the two bounds. Appendix E.5 gives the setwise comparison and a counterexample to necessity.

6.3 Public power, informational rents, and type incidence

For each linked private record, define the agenda-game informational rent type by type,

$$IR_g^A(o) = V_g^A(o) - v_g^A(o), \quad \Delta IR^A(o) = IR_U^A(o) - IR_M^A(o).$$

Because every institutional contrast now uses the orientation unanimity minus majority, the decomposition is

$$\boxed{\Delta V^A(o) = \Delta v^A(o) + \Delta IR^A(o)}, \quad \Delta IR^A(o) = \Delta V^A(o) - \Delta v^A(o).$$

The same identity holds ex ante after applying the affine map to the same linked vector.

Proposition 6.3 (Informational-rent incidence under unanimity). *In every existing unanimity fiber, $IR_U^A(\ell) \geq 0$, $IR_U^A(h) \leq 0$, and at least one inequality is strict.*

Positive informational rent belongs to the low type, which can receive a concession designed for the high type. The high type receives no such premium and may lose relative to its public benchmark. Appendix E.9 reports the exact cellwise translations, and Appendix B.8 proves the underlying exhaustive unanimity correspondence. When $\Delta v^A(\ell) < 0$, public information favors majority for the low type. If a comparable private member nevertheless has $\Delta V^A(\ell) > 0$, then member by member

$$\Delta IR^A(\ell) > -\Delta v^A(\ell) > 0.$$

Information is then the only component pointing toward unanimity and more than offsets majority's public advantage. This statement does not select a member or imply a global sign for $\Delta IR^A(\ell)$.

Agenda also changes the informational rent relative to the no-agenda benchmark. Let IR_g^B be the frozen Round-1 correspondence from the baseline. Its date- A value is βIR_g^B , so

$$I_g = IR_g^A - \beta IR_g^B, \quad \Delta I = \Delta IR^A - \beta \Delta IR^B.$$

This is the only multiplication by β in the interaction. Under unanimity, agenda weakly reduces informational rent in every existing high-prior fiber; under majority and across rules the interaction

remains set-valued wherever its sources do.

6.4 The structural effect of the agenda stage

The agenda effect compares presence and absence of the mandatory date- A stage within a fixed information regime. All four arms in Figure 6 are expressed at date A .

	Baseline without hegemonic agenda B	Hegemonic agenda A
Public type	$\beta v_g^B(o)$	$v_g^A(o)$
Private type	$\beta V_g^B(o)$	$V_g^A(o)$

$$\underbrace{T_g(o)}_{\text{agenda under private information}} = \underbrace{D_g(o)}_{\text{agenda under public information}} + \underbrace{I_g(o)}_{\text{agenda-information interaction}}.$$

Figure 6: The factorial agenda-information design. The treatment adds an earlier, mandatory proposal stage; the control begins in Round 1 and is discounted once when evaluated at date A .

Formally,

$$D_g(o) = v_g^A(o) - \beta v_g^B(o), \quad T_g(o) = V_g^A(o) - \beta V_g^B(o),$$

Proposition 6.4 (Agenda-effect decomposition). *For every complete record in which both arms exist,*

$$\boxed{T_g(o) = D_g(o) + I_g(o)}.$$

Under unanimity, T_U weakly benefits both types wherever both arms exist.

Appendix E.11 derives the identity by substituting the informational-rent definition in both arms. Appendix E.13 reports the exact unanimity cells. This is a structural causal contrast inside the model, not an empirically identified estimand. Under multiplicity it is a correspondence of admissible contrasts.

The public effects are

$$D_U(o) = 1 - \beta$$

and

$$D_M(o) = \begin{cases} v_M^{\text{safe}} - \frac{e}{m}\beta^2 o, & o \leq 1/m, \\ v_M^{\text{safe}} - \beta o, & 1/m < o < o_M^*, \\ 0, & o \geq o_M^*. \end{cases}$$

Thus agenda never lowers H 's public payoff under majority, but it can have no direct value when the mandatory proposal induces the same delayed continuation as the control. The first branch applies at $o = 1/m$; the frozen continuation selection creates a downward jump in D_M immediately to its right. The threshold o_M^* can lie outside $o \in (0, 1)$.

Under private majority,

$$T_M = D_M + I_M$$

is the exact translation of the interaction correspondence by the fixed public vector. No general sign is imposed. Across rules, member by member within a complete admissible product,

$$\Delta D(o) = D_U(o) - D_M(o), \quad \Delta T(o) = T_U(o) - T_M(o) = \Delta D(o) + \Delta I(o).$$

The diagonal object $Q_g = v_g^A - \beta V_g^B$ changes agenda and information simultaneously. It is useful for comparison but is not a one-factor causal effect; Appendix E reports it separately.

7 Discussion

7.1 Pivotality, substitutes, and contested strength

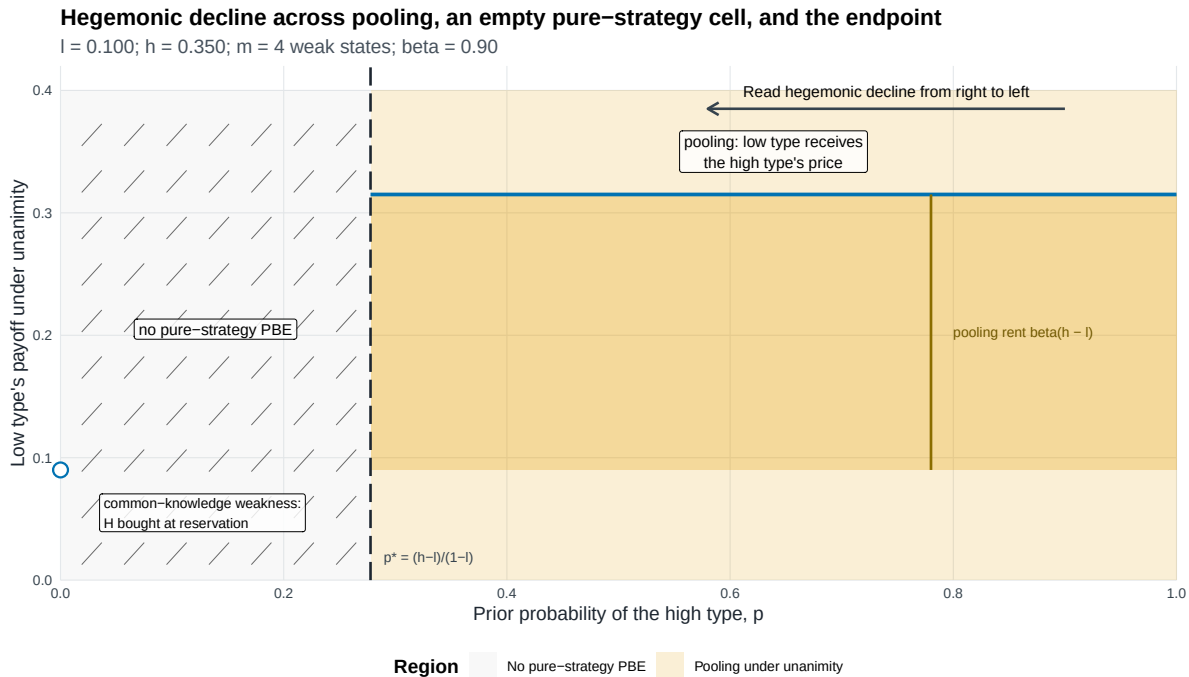
The model's central distinction is technological. Under majority, the proposer can replace the hegemon with another uninformed weak-state vote. Under unanimity, the hegemon's approval is an essential input. Removing the substitute changes both bargaining power and the way private information enters proposals.

The complete-information benchmark shows the power component. If the known hegemon is cheap enough, majority includes it and both rules pay the same discounted threshold. If it is expensive, majority excludes it while unanimity must still buy its vote. The informational comparison asks a

different question: after netting out that public-type difference, how much does private information change each rule's payoff?

The answer is not simply that private information always benefits a hegemon. Pooling under unanimity benefits the low type because weak states price it as high. Screening under majority can remove that rent, while exclusion can make the comparison depend on whether the high threshold discounted to the present exceeds the low current disagreement payoff. The high type's rent changes only when its treatment differs across the private and public majority games.

The empty middle-belief cell is equally important for scope. It records the failure of every pure voting pattern under the declared belief and pivotality conditions. It does not supply a payoff for institutional comparison and does not justify filling the gap by interpolation.



The figure should be read from high to low p . Above p^* , pooling pays the low type βh , creating the rent $\beta(h-l)$ relative to its reservation price βl . The light orange field identifies the pooling region; the darker band isolates the payoff-relevant rent between βl and βh . For $0 < p \leq p^*$, the hatched area records the absence of a perfect Bayesian equilibrium in pure ballot strategies; $p = 0$ is an isolated complete-information endpoint, not an interval, and the low type is bought at reservation. The figure makes no claim about equilibria involving mixed ballot strategies. Note: Model-generated regions; all boundaries closed-form.

Figure 7: A schematic movement in beliefs about hegemonic strength. The hatched region is the set in which the unanimity game has no equilibrium in pure ballot strategies under the maintained concept. Historical annotations, if used, are illustrations rather than empirical tests or calibration.

7.2 WTO creation, exit, and observable implications

The close of the Uruguay Round provides a sharper institutional illustration than OPEC. Steinberg reports that the United States and the European Community backed the single undertaking and, after joining the WTO, withdrew from GATT 1947 (Steinberg 2002). Their withdrawal terminated the old GATT obligations, including most-favored-nation treatment, toward countries that did not accept the Final Act and join the new organization. The maneuver made the preexisting trading relationship a less valuable fallback and helped give the WTO package broad membership.

The two parts of that strategy map to different bargaining objects. The single undertaking joined the Uruguay Round agreements into one package and restricted selective acceptance. Withdrawal from GATT 1947 changed the value of rejecting that package. The latter supplies a concrete micro-foundation for disagreement payoffs: market power permits a major state to shape what agreement failure is worth to itself and to its partners. If the precise value of the major state's outside option is not common knowledge, bargaining over an approval that cannot be replaced also creates the screening problem analyzed in the model.

The episode is not a literal implementation of the baseline. Steinberg shows that the United States and European Community exercised substantial de facto agenda influence, whereas the benchmark denies the hegemon an exclusive formal proposal right. The benchmark isolates one channel inside a historically richer process. It asks what the consensus requirement contributes after agenda control is held fixed, not whether agenda control was absent or whether the model identifies why the United States created the WTO.

This distinction yields observable implications. Informational power through pivotality should be strongest where a major state's approval has no coalition substitute, its fallback is hard for partners to price, and the negotiated package can vary enough to compensate uncertain participation constraints. In the pooling region, the signature is a smooth agreement that grants a concession suited to a stronger possible type. A theory of signaling through rejection instead predicts visible delay or failed proposals before beliefs change. Under majority, the availability of a substitute coalition should reduce the return on private information even when the underlying outside option remains unequal.

Steinberg also emphasizes information flowing in the opposite direction: the consensus process helps powerful agenda setters elicit weaker states' preferences. The model studies weak proposers who are uncertain about the hegemon's continuation value. An empirical application must distinguish these directions of information and use language of consistency rather than treat the WTO episode as identification of the formal mechanism.

7.3 Limits

The baseline fixes the voting rule, keeps the institutional pie at one, gives proposal power only to weak states, and uses two types and two rounds. The agenda extension relaxes only that baseline proposal restriction by adding the earlier mandatory proposal stage for H . Neither part analyzes endogenous rule choice or an endogenous agenda protocol; the paper also does not analyze more than two types, sequential roll-call voting, or intrinsic agreement benefits. The restriction $m \geq 3$ excludes the three-player case. The strict discounting assumption excludes the corner $\beta = 1$, where additional indifferences would require a separate analysis.

The endpoint-support condition also matters. A type assigned zero probability by the prior cannot be resurrected after an off-path action. This makes the degenerate-prior cells coincide with their complete-information games. In the interior, both types remain in the prior's support and off-path actions by H can sustain a range of posterior beliefs.

The agenda extension preserves rather than resolves equilibrium multiplicity. Its majority effects, institutional comparisons, and some interactions are correspondences. Every robust sign is therefore setwise, and every other sign is member-specific. Empty cells record absence of a pure PBE under M/S/B, not a zero effect. The structural contrast T also depends on the maintained date convention and mandatory proposal stage; it is distinct from the diagonal comparison Q .

Finally, the WTO discussion is a theoretical application. The model neither identifies why the United States and European Community designed the WTO nor establishes that a particular historical actor was a low or high type. It also abstracts from plurilateral agreements, subsidiary voting procedures, ratification, implementation, and repeated-game reputation. Those mechanisms may complement or substitute for the channels isolated here.

8 Conclusion

Consensus can benefit a hegemon even when weak states control the agenda because unanimity removes substitutes for the hegemon's vote. Complete information reveals the familiar pivotality component. Private information adds a distinct component: when unanimity pools at the high threshold, a low type can earn rent; whether this exceeds its majority rent depends on the majority correspondence and the outside-option region.

Restoring hegemonic agenda power does not erase the benchmark result. It separates real power from its informational shadow. When the outside option is high, the public component can itself favor unanimity. When public information favors majority, a private unanimity ranking can arise only if the relative informational rent offsets that public disadvantage. The structural effect of the agenda stage is $T = D + I$: its direct value and its interaction with information are distinct, and neither has a universal institutional sign.

Formal equality of votes can therefore coexist with unequal bargaining power through three channels: outside options, proposal rights, and the price of an approval with no substitute. The WTO application is consistent with their coexistence, while the identifying benchmark shows that the last channel does not require an exclusive formal proposal right. The paper's conclusions remain conditional on complete linked correspondences, their domains, and the cells in which the maintained equilibrium concept exists.

Appendix A: Protocol, beliefs, and incentives

A.1 Complete transition and payoff rules

For a proposal $x = (x_H, x_1, \dots, x_m)$, let $a_H \in \{Y, N\}$ and let the weak responders' votes be a_{-i} . The proposer votes yes. If the quota is met, the proposal is implemented among the parties. Each weak respondent receives x_j , the proposer receives x_i , and H receives x_H after yes; after no, H receives o and x_H is paid to no one. If the quota is not met in Round 1, the public vote vector leads to Round 2; every continuation payoff is then multiplied by β . If the quota is not met in Round 2, weak states receive zero and H receives o .

Agreement and disagreement payoffs are mutually exclusive at every history: no history pays a player both an allocation from the pie and an outside option. All equilibrium exclusions pay H exactly o .

A.2 Beliefs and ballot restrictions

Uninformed weak-state proposals and votes preserve the current belief. Structural consistency is the following restriction. Fix a profile. At each ballot, the entering belief is the current posterior, and H 's prescribed law is the type-contingent vote distribution the profile assigns at that ballot. The posterior after any vote vector depends on the history only through the entering belief, the prescribed law, and H 's realized vote; it is therefore invariant across vote vectors of the same ballot that differ only in weak-state votes. If H 's realized vote has positive probability under the prescribed law and the entering belief, the posterior is the Bayes update given that law, including at ballots reached by earlier weak-state deviations. If the Bayes denominator is zero, the posterior is a single free value attached to that ballot-and-vote pair, chosen within the support of the prior: any value in $[0, 1]$ for an interior prior, and the degenerate posterior at $p = 0$ or $p = 1$. Distinct ballots may carry distinct free values. This condition is a declared restriction of our equilibrium concept; it is not the consistency notion of sequential equilibrium (Kreps and Wilson 1982), and it is narrower than the never-dissuaded discipline of Osborne and Rubinstein (1990).

A weak responder compares its allocation with the continuation that would follow if its vote switched passage into failure. It votes yes at equality. H maximizes its type-contingent con-

tinuation payoff and votes yes at exact indifference. These restrictions apply at every proposal, including off-path proposals.

Appendix B: Proofs

B.1 Proof of Proposition 5.1

In terminal majority, a weak responder's disagreement value is zero, so any nonnegative allocation induces yes under the indifference-to-yes convention. There are $m - 1$ weak responders, and $k = \lfloor (m + 1)/2 \rfloor \leq m - 1$ for $m \geq 3$; their votes can therefore pass the proposal without H . At any such proposal, a nonpivotal H votes yes exactly when $x_H \geq o$: yes yields x_H , whereas no yields o . Yet every $x_H > 0$ is strictly suboptimal for the proposer. If H votes no, x_H is paid to no one; if H votes yes, it is paid for a vote that is unnecessary for passage. In either case, setting $x_H = 0$ and assigning that amount to the proposer preserves every weak responder's allocation and ballot, preserves passage, and raises the proposer's payoff by x_H . Positive payments to weak responders can likewise be reduced to zero without changing their ballots. Hence the unique equilibrium outcome has $x_H = 0$, zero payments to every weak responder, the proposer keeps the unit pie, and H votes no and receives o .

In terminal unanimity, every weak responder accepts zero and H accepts exactly when $x_H \geq o$. The proposer therefore sets $x_H = o$.

Move to Round 1. The terminal continuation of a weak state before recognition under majority is $1/m$, hence $w = \beta/m$. The continuation of H is βo . Under unanimity, paying H exactly βo and each weak responder its continuation implements immediate agreement and leaves the proposer strictly more than delay because $\beta < 1$.

Under majority, inclusion costs $(k - 1)w + \beta o$; exclusion costs kw . Inclusion is weakly cheaper exactly when $o \leq 1/m$. At equality the proposer is indifferent, and the proposal tie-break selects inclusion because H receives $\beta o < o$. This gives the stated payoffs. $\square$

B.2 Proof of Proposition 5.2

The majority argument is the terminal-majority part of the preceding proof. Before recognition, symmetry gives each weak state expected payoff $1/m$.

Under unanimity, weak responders accept zero. Any accepted offer to H can be reduced to the lowest threshold consistent with its acceptance set. The only nonempty acceptance sets are the low type alone, implemented by $x_H = \ell$, and both types, implemented by $x_H = h$. The first gives the proposer $(1 - p)(1 - \ell)$; the second gives $1 - h$. The low offer is weakly preferred exactly when

$$p \leq \frac{h - \ell}{1 - \ell} = p^*.$$

At equality the low offer gives H the lower expected payoff and is selected. $\square$

B.3 Proof of Proposition 5.3

A responding weak state votes yes exactly when $x_j \geq w = \beta/m$. This threshold is independent of the posterior and of H 's vote because a weak state's terminal-majority continuation is always $1/m$. Let n_Y be the number of such responders. If $n_Y \geq k$, the weak-state votes pass the proposal without H . A nonpivotal type with outside option o then votes yes exactly when $x_H \geq o$: yes yields x_H , whereas no yields o . Nevertheless, every $x_H > 0$ in this class is strictly suboptimal for the proposer. Moving x_H to the proposer's own allocation preserves the weak-state votes and passage and raises the proposer's payoff by x_H . Hence the optimal exclusion candidate sets $x_H = 0$; both types then vote no and receive their outside options. If $n_Y = k - 1$, H is pivotal and a type with outside option o votes yes exactly when $x_H \geq \beta o$, because no leads to the terminal-majority continuation. If $n_Y \leq k - 2$, the proposal fails regardless of H 's vote; the two ballot actions deliver the same terminal-majority continuation, so the indifference-to-yes convention selects yes.

The preceding dominance argument leaves only $x_H = 0$ in the class $n_Y \geq k$. In the pivotal class $n_Y = k - 1$, weak payments can be reduced to w and x_H to $\beta\ell$ or βh , according to the desired acceptance set; proposals in the remaining class deliver delay. Assigning the remaining pie to the proposer weakly increases its payoff. This leaves four candidates: exclusion with payoff Π_E , screening with payoff $\Pi_S(p)$, pooling with payoff Π_P , and deliberate delay with payoff w .

Exclusion is always feasible and

$$\Pi_E - w = 1 - \frac{\beta(k+1)}{m} > 0,$$

so delay is never selected. A screening or pooling candidate that could beat exclusion is automatically feasible.

The decisive payoff differences are

$$\Pi_P - \Pi_E = \beta(1/m - h)$$

and

$$\Pi_S(p) - \Pi_E = (1-p)\beta(1/m - \ell) - p(1 - \beta(k+1)/m).$$

Where pooling can beat exclusion, comparing screening with pooling yields $p_{S=P}$; comparing screening with exclusion yields $p_{S=E}$. Substituting the signs of $\ell - 1/m$ and $h - 1/m$ produces the five cases in Proposition 5.3.

At screening ties, its expected payoff to H is strictly smaller than that of the tied alternative, so the proposal tie-break selects screening. When $h = 1/m$, exclusion and pooling give the proposer the same payoff. Their payoffs to H determine the stated selection; exact equality leaves their entire proposal segment. All selected proposals exhaust the pie. Finally, permuting the identities of weak responders does not change any comparison, which establishes the reported identity multiplicity. $\square$

B.4 Proof of Proposition 5.4

Transport the terminal unanimity continuation into Round-1 units. If the posterior is μ , a weak state's continuation is

$$c_U(\mu) = \begin{cases} (1-\mu)w_\ell^U, & \mu \leq p^*, \\ w_h^U, & \mu > p^*. \end{cases}$$

Thus $c_U(\mu) \in [w_h^U, w_\ell^U]$, and $x_j = w_\ell^U$ forces every weak responder to vote yes under any admissible posterior.

At $p = 0$, support preservation fixes every posterior at zero. The proposal

$$x_H = \beta\ell, \quad x_j = w_\ell^U, \quad x_i = 1 - \beta\ell - (m-1)w_\ell^U = w_\ell^U + 1 - \beta$$

is accepted. Each responder and H is exactly indifferent and votes yes. Any lower payment loses a required vote; any higher payment reduces the proposer's residual. A delay gives the proposer w_ℓ^U , so immediate agreement is strictly better by $1 - \beta$.

For $p > p^*$, the analogous pooling proposal is

$$x_H = \beta h, \quad x_j = w_h^U, \quad x_i = 1 - \beta h - (m-1)w_h^U = w_h^U + 1 - \beta.$$

Both types of H and every weak responder accept, and immediate agreement again exceeds delay by $1 - \beta$.

For completeness, write $u = \min_j x_j$ after an arbitrary proposal and order the strategy of H as low-type/high-type. At $p = 0$, posterior support is always the low type. The profile (Y, Y) is sequentially rational if $u < w_\ell^U$, or if $u \geq w_\ell^U$ and $x_H \geq \beta h$; (N, N) is rational if $u \geq w_\ell^U$ and $x_H < \beta\ell$; and (Y, N) is rational if $u \geq w_\ell^U$ and $\beta\ell \leq x_H < \beta h$. The profile (N, Y) is never rational. These cases are mutually exclusive and exhaustive.

For $p^* < p < 1$, the profile (Y, Y) is rational if $u < w_h^U$, or if $u \geq w_h^U$ and $x_H \geq \beta h$; (N, N) is rational if $u \geq w_h^U$ and $x_H < \beta h$; and neither separating profile is rational. Bayes gives the current prior after an on-profile pooled action. When yes is a zero-probability action, an admissible posterior μ_Y supports all weak yes votes exactly when $c_U(\mu_Y) \leq u$, which is possible in the stated no profile because $u \geq w_h^U$. At $p = 1$, support preservation fixes the posterior at one and the same two pooled-profile conditions apply. Thus every proposal in both existence domains has a pure, sequentially rational response completion.

It remains to show the empty cell. For $0 < p \leq p^*$, consider

$$s^\dagger : \quad x_H = \beta\ell, \quad x_j = w_\ell^U \text{ for every weak responder,} \quad x_i = 1 - \beta\ell - (m-1)w_\ell^U.$$

All weak responders must vote yes. Enumerate the four pure type-contingent profiles for H , ordered

low/high:

1. Under (Y, Y) , the high type receives $\beta\ell < \beta h$ and profitably votes no to obtain continuation βh .
2. Under (N, N) , the low type's no gives $\beta\ell$, so the indifference-to-yes convention requires yes.
3. Under (Y, N) , the low type can imitate the high type's no. That action reveals the high type and yields continuation $\beta h > \beta\ell$.
4. Under (N, Y) , the high type can imitate the low type's no and obtain continuation βh , while the prescribed comparison cannot support strict no against yes at the offer.

No pure profile is sequentially rational after $s^\dagger$. Because a PBE must specify a valid pure response after every feasible proposal, no pure-ballot PBE exists in this cell. This also explains the discontinuity: at $p = 0$, support preservation rules out the posterior that drives the imitation argument; for every positive p , both types are possible. $\square$

B.5 Proof of Proposition 5.5

The private majority payoff vectors for H are

$$V_M^{B,S} = (\beta\ell, \beta h), \quad V_M^{B,P} = (\beta h, \beta h), \quad V_M^{B,E} = (\ell, h).$$

The private unanimity vectors are $(\beta\ell, \beta h)$ at $p = 0$, empty for $0 < p \leq p^*$, and $(\beta h, \beta h)$ for $p > p^*$. Componentwise subtraction gives the proposition. On any proposal segment, subtraction is affine in the common proposal weight, so the payoff set and outcome set remain atomic. $\square$

B.6 Proofs of Propositions 5.6 and 5.7

From Proposition 5.1, the public payoff vectors are

$$v_M^B = (v_M^B(\ell), v_M^B(h)), \quad v_U^B = (\beta\ell, \beta h).$$

Subtracting these from each private vector gives the table in Proposition 5.6. In particular, unanimity is payoff-equivalent to its public endpoint at $p = 0$, is empty in the middle cell, and pools above the cutoff, which yields $(\beta(h - \ell), 0)$.

Subtract IR_M^B from IR_U^B . In the high-belief cell, screening gives $(\beta(h - \ell), 0)$ when both types are publicly included and $(\beta(h - \ell), (1 - \beta)h)$ when only the low type is publicly included; pooling gives zero. Exclusion gives $(\beta h - \ell, -(1 - \beta)h)$, $(\beta h - \ell, 0)$, or $(\beta(h - \ell), 0)$, respectively when both types are included, only the low type is included, or both types are excluded. The endpoint follows by the same calculation. An empty source set remains empty. For the residual segment, affine subtraction with its common weight λ yields the one-dimensional segment stated in the propositions, not the Cartesian product of marginal envelopes. $\square$

B.7 Proof of the private-majority agenda correspondence

Fix the primitives and the anonymous Markov continuation selection χ from Appendix E. For posterior μ , let $c_\chi(\mu)$ be a weak state's continuation payoff and $h_{\chi,o}(\mu)$ the continuation payoff of type o , both in the native units of the baseline. Transporting the continuation to date A gives

$$r_\chi(\mu) = \beta c_\chi(\mu), \quad d_{\chi,o}(\mu) = \beta h_{\chi,o}(\mu).$$

Every baseline branch gives $0 < c_\chi(\mu) \leq 1/m$. Hence

$$0 < r_\chi(\mu) \leq \frac{\beta}{m}, \quad kr_\chi(\mu) < 1.$$

As-if-pivotal voting therefore implies that weak state j votes yes if and only if $x_j \geq r_\chi(\mu)$. The equality case passes by the declared indifference rule. A proposal passes exactly when at least k weak states receive that amount. Conditional on μ , the best passing proposal pays $r_\chi(\mu)$ to exactly k weak states and leaves

$$a_\chi^{\text{pass}}(\mu) = 1 - kr_\chi(\mu)$$

to H . A proposal such as $(1, 0, \dots, 0)$ is rejected and attains $d_{\chi,o}(\mu)$. Thus both terms in

$$\max\{a_\chi^{\text{pass}}(\mu), d_{\chi,o}(\mu)\}$$

are attainable at every fixed posterior.

For a pure interior profile, the public support contains at most two proposals. Its complement is dense in the proposal simplex. If the canonical best passing proposal at μ^{off} is itself on the support, reduce H 's share by $\varepsilon > 0$, keep the k threshold payments fixed, and choose a sequence $\varepsilon \downarrow 0$ that avoids the finitely many supported proposals. Rejected proposals are also available off support. Consequently, the supremum payoff from an unsupported proposal is exactly

$$O_o(\rho) = \max\{a_\chi^{\text{pass}}(\mu^{\text{off}}), d_{\chi,o}(\mu^{\text{off}})\}.$$

Because rejection is weakly more valuable to the high type while the passing term is common, $O_h(\rho) \geq O_\ell(\rho)$.

We can now enumerate the pure profiles. Under pooling, the posterior is p . If the common proposal passes, both types receive the same share z ; feasibility requires $z \leq a_p^{\text{pass}}$, and absence of unsupported deviations requires $z \geq O_h(\rho)$. This gives the first row of Table 7. If the pooled proposal is rejected, type o receives $d_{o,p}$, so the two inequalities in the pooling-delay row are necessary and sufficient.

Under separation, Bayes' rule assigns posterior zero to the low-type proposal and posterior one to the high-type proposal. If both pass, either type can copy the other's proposal and receive the same share. Bilateral imitation therefore forces a common z . Feasibility and unsupported deviations give

$$O_h(\rho) \leq z \leq \min\{a_0^{\text{pass}}, a_1^{\text{pass}}\}.$$

If the low type passes and the high type delays, imitation, feasibility, and unsupported deviations require

$$\max\{d_{\ell,1}, O_\ell(\rho)\} \leq z \leq \min\{a_0^{\text{pass}}, d_{h,1}\}, \quad d_{h,1} \geq O_h(\rho).$$

These are exactly the fourth row of the table. The reverse pattern is impossible: it would require $d_{\ell,0} \geq z \geq d_{h,0}$, although $d_{h,0} > d_{\ell,0}$. Finally, if both proposals are rejected, bilateral imitation gives $d_{\ell,0} \geq d_{\ell,1}$ and $d_{h,1} \geq d_{h,0}$; the remaining two inequalities in the last row exclude unsupported deviations. The pooling/separating and pass/reject enumeration is exhaustive. Canonical

passing proposals and clearly rejected proposals attain every listed payoff, proving sufficiency as well as necessity.

For completeness with arbitrary Borel proposal laws, set $S_M = \text{supp}(\bar{\sigma})$ and define

$$u_o(x) = b_M(x)x_H + \{1 - b_M(x)\}d_{\chi,o}(\pi(x)), \quad V_o = \int u_o(x) \sigma_o(dx).$$

Let u_o^{off} evaluate an unsupported proposal at μ^{off} , and put

$$\mathfrak{D}_o(S_M) = \sup_{x \notin S_M} u_o^{\text{off}}(x),$$

with the supremum of the empty set equal to $-\infty$. The proposal laws generate a PBE if and only if, in addition to Bayes, the ballot rule, and the continuation selection in R_M ,

$$\begin{aligned} u_o(x) &\leq V_o && \text{for every } x \in S_M, \\ u_o(x) &= V_o && \text{for } \sigma_o\text{-almost every } x, \\ V_o &\geq \mathfrak{D}_o(S_M), && o \in \{\ell, h\}. \end{aligned}$$

Necessity follows because each type can copy every supported proposal and choose any unsupported proposal, while a mixed strategy assigns probability one to best responses. Conversely, prescribe the cutoff ballots and the continuation $\chi(\pi(x))$. The first line rules out profitable supported deviations, the third rules out unsupported ones, and the second restricts each type's law to best responses. This proves sequential rationality at every proposal. The local Bayes ratio is a pointwise limit of Borel ratios on the closed support and is constant off it; hence π , the finite cutoff ballot, and both payoff maps are Borel. The conditions are therefore also sufficient for the mixed correspondence stated in Appendix E.2.

At $p \in \{0, 1\}$, prior support fixes every posterior at the endpoint. Each type's complete Borel best-response correspondence is consequently the set of probability measures supported on its argmax between passage and rejection, with value $\bar{V}_o$.

It remains to prove existence and the lower bound used in Appendix E.5. Let

$$v_M^{\text{safe}} = 1 - \frac{k\beta}{m}, \quad T = \frac{v_M^{\text{safe}}}{\beta} = \frac{1}{\beta} - \frac{k}{m} > \frac{1}{m}.$$

If $h \leq T$, take $\rho = 1$ and pool on the canonical passing proposal at posterior p . To see that rejection is not profitable, consider the three possible continuation outcomes. Exclusion gives the high type at most $\beta h \leq \beta T = v_M^{\text{safe}}$; screening or pooling gives it at most $\beta^2 h < v_M^{\text{safe}}$, while the best passing proposal at that posterior leaves at least v_M^{safe} . The same inequalities hold for the low type and are preserved on the residual continuation segment by convexity.

If $\ell \leq T \leq h$, take $\mu^{\text{off}} = 1$. At posterior one, the majority continuation excludes H , so

$$d_{\ell,1} = \beta\ell \leq v_M^{\text{safe}} \leq \beta h = d_{h,1}.$$

At posterior zero, the best passing proposal leaves at least v_M^{safe} . Thus the low type passes with that share and the high type rejects; the displayed inequalities rule out imitation in both directions and the posterior-one calculation also rules out unsupported deviations. If $T \leq \ell$, then $T > 1/m$ implies that the majority continuation excludes H at every posterior. Rejection gives each type $\beta o \geq v_M^{\text{safe}}$, so both types can pool on delay for any ρ . The adjacent constructions remain valid at equality; endpoints follow from the preceding argmax description. Thus a majority PBE exists for every admissible primitive and prior for some ρ .

Moreover, at any posterior H can pay β/m to any k weak states and retain v_M^{safe} . Since every cutoff is at most β/m , this proposal passes regardless of the belief it induces. A clearly rejected proposal gives type o at least $\beta^2 o$. Hence every majority equilibrium satisfies

$$V_M^A(o) \geq \max\{v_M^{\text{safe}}, \beta^2 o\},$$

which establishes the lower bound used by Proposition 6.2. $\square$

B.8 Proof of the private-unanimity agenda correspondence

Write

$$z_L = v_U^A(\ell) = 1 - \beta + \beta^2\ell, \quad z_H = v_U^A(h) = 1 - \beta + \beta^2h, \quad d = \beta^2h, \quad \Delta = z_L - d.$$

At posterior zero, unanimity prices every weak vote at $r_U(\ell)$; at a posterior above p^* , it prices every weak vote at $r_U(h)$. Since one no vote rejects the proposal, sequential rationality after that vote requires a continuation at the same posterior. The baseline unanimity correspondence has no member on $(0, p^*]$. Therefore every disciplined posterior and the common off-support posterior must lie in $\mathcal{P}_C = \{0\} \cup (p^*, 1]$.

The cutoff sets are closed because equality is accepted. The maximal shares that H can retain under the two admissible prices are attained at x^ℓ and x^h , and equal z_L and z_H , respectively. Rejection gives the high type d at every admissible posterior. It gives the low type $\beta^2\ell$ at posterior zero and d at a high posterior. In particular,

$$0 < r_U(h) < r_U(\ell), \quad z_H = z_L + \beta^2(h - \ell) > z_L, \quad z_H > d.$$

We first establish two reduction lemmas. For an interior prior, the two types receive the same equilibrium payoff. A signal used by the high type has a high posterior. If it passes, the low type can copy it and obtain the same share; if it rejects, both types receive d . A positive-mass zero-posterior signal can be used only by the low type. It cannot reject in equilibrium: the high type can guarantee d , and the low type can copy a best high-type signal, whereas rejection at zero gives it only $\beta^2\ell < d$. It must therefore pass, after which the high type can copy it and obtain the same share. These bilateral imitation inequalities, together with indifference on the support of each mixed strategy, imply a common payoff V .

Next let

$$\lambda_0 = \int 1\{\mu(x) = 0\} d\bar{\sigma}(x)$$

be the public mass of zero-posterior proposals. If $\lambda_0 > 0$, the low type uses such a proposal and can receive at most z_L ; only x^ℓ attains that maximum. The common-payoff result and the high

type's rejection option require $z_L \geq d$, or $\Delta \geq 0$. An off-support high posterior would permit the deviation x^h , which yields $z_H > z_L$; hence $\mu^{\text{off}} = 0$. It follows that $V = z_L$, $\sigma_\ell(\{x^\ell\}) > 0$, and $\sigma_h(\{x^\ell\}) = 0$. If instead $\lambda_0 = 0$, Bayes plausibility gives

$$p = \int \mu(x) d\bar{\sigma}(x) > p^*,$$

because the posterior is above p^* almost surely. These two cases are exhaustive.

Suppose first that $\lambda_0 > 0$. The preceding argument gives the domain of the low-posterior family. Every used high-posterior proposal must either pass with H 's share z_L , or, only when $z_L = d$, reject and deliver the same payoff. The pointwise deviation restrictions in Appendix E.3 are necessary because either type can choose every proposal, including supported limit points of zero mass. They are sufficient because all used proposals then yield z_L , while every other disciplined or unsupported proposal yields at most z_L . A pure witness is

$$\sigma_\ell = \delta_{x^\ell}, \quad \sigma_h = \delta_{x^S}, \quad x^S = (z_L, r_U(h), \dots, r_U(h)).$$

The proposal x^S passes and is feasible because $z_L + mr_U(h) = 1 - \beta^2(h - \ell) < 1$. Neither type gains by imitation or by rejection precisely when $\Delta \geq 0$. This proves necessity and sufficiency of the low-posterior family.

Now suppose $\lambda_0 = 0$, so $p > p^*$. If $\mu^{\text{off}} = 0$, the common payoff must satisfy

$$\max\{z_L, d\} \leq V \leq z_H.$$

The lower bound excludes the best zero-posterior passing and rejection deviations; the upper bound is the maximal feasible share at a high posterior. For every V in the interval, the pooling proposal

$$x(V) = \left(V, \frac{1-V}{m}, \dots, \frac{1-V}{m} \right)$$

passes because $V \leq z_H$ implies $(1-V)/m \geq r_U(h)$. It gives posterior p and leaves no profitable off-support deviation because $V \geq \max\{z_L, d\}$. This proves existence of every payoff in the high-posterior family with zero off-support belief; the pointwise restrictions in Appendix E.3 give the

corresponding necessary and sufficient condition for arbitrary Borel laws.

If $\mu^{\text{off}} > p^*$, the off-support proposal x^h guarantees z_H . Feasibility also bounds the common payoff by z_H ; hence $V = z_H$. The only passing proposal that leaves this share is x^h , so both proposal laws are δ_{x^h} . An off-support posterior in $(0, p^*]$ is inadmissible because it has no baseline continuation. Thus the two high-posterior subfamilies in Appendix E.3 are exhaustive. Combining them with the dichotomy for λ_0 proves that there is no third interior family and gives all stated existence cells.

At $p = 0$, support preservation fixes every posterior at zero. The low type uniquely chooses x^ℓ . The counterfactual high type chooses x^ℓ if $z_L > d$, may mix x^ℓ with rejected proposals if $z_L = d$, and chooses any rejected best response if $z_L < d$. This gives the endpoint correspondence in Appendix E.3. At $p = 1$, every posterior is one and both types uniquely choose x^h , since $z_H = d + 1 - \beta > d$. Finally, any passing unanimity proposal gives every weak state at least $r_U(h)$, so it leaves H at most z_H ; every rejection payoff is at most $d < z_H$. Therefore every unanimity member satisfies

$$V_U^A(o) \leq z_H = v_U^A(h).$$

Together with the majority lower bound in Appendix B.7, this proves the two bounds used in Appendix E.5. The linked payoff image, informational-rent incidence, and total agenda-effect cells then follow by the fixed-vector subtractions displayed in Appendices E.9 and E.13. $\square$

B.9 Proof of the exact and economic representations

The argument is the same for majority and unanimity, so fix a rule g . Let Z_g be the realized-record space defined in Appendices E–F. It is a finite product and disjoint union of compact Polish proposal, posterior, ballot, continuation-label, and terminal-record spaces; hence it is itself compact Polish. A common permutation $\tau \in G = S_m$ of weak-state names acts on Z_g by a homeomorphism T_τ . Relabeling a PBE by T_τ preserves feasibility and payoffs, transports relative balls and supports, commutes with the local Bayes limit, and maps ballots, continuation kernels, and deviations bijectively to payoff-equivalent objects. The relabeled object is therefore also a PBE.

Set $X_g = \mathcal{P}(Z_g)^2$, with the weak topology, and let the finite group act diagonally on the pair of type-specific realized laws. This is a Polish space with a continuous group action. For $x \in X_g$,

define the orbit law

$$\Lambda_x = \frac{1}{|G|} \sum_{\tau \in G} \delta_{\tau x}.$$

For every Borel set $C \subseteq X_g$,

$$\Lambda_x(C) = \frac{1}{|G|} \sum_{\tau \in G} 1_C(\tau x)$$

is Borel in x , so $x \mapsto \Lambda_x$ is Borel. Right multiplication only reorders the finite sum, proving invariance. If $x' = \tau x$, the two orbit laws coincide. Conversely, if $\Lambda_x = \Lambda_{x'}$, the Borel singleton $\{x'\}$ has positive mass under $\Lambda_{x'}$, hence also under Λ_x ; therefore $x' = \tau x$ for some τ . The orbit law is thus a complete invariant of diagonal relabeling, which proves the claim about Sig_g^{ex} .

For the economic layer, choose a Borel injection of Z_g into $[0, 1]$ and let q_g map each record to the least element of its finite orbit. This gives a Borel transversal and a Borel quotient map satisfying $q_g \circ T_\tau = q_g$. If $f : Z_g \rightarrow E$ is Borel and invariant, define $\bar{f}$ on the transversal by its value at the selected representative. Invariance makes the definition independent of the original named record, and surjectivity of q_g gives uniqueness. Thus

$$f = \bar{f} \circ q_g, \quad \int f d\Gamma_o = \int \bar{f} d(q_g)_\# \Gamma_o.$$

Payoffs, passage and delay, the posterior law, anonymous terminal outcomes, and the ordered multiset of weak-state payoffs are invariant Borel maps, so they factor through the economic summary. The same is true of the payoff and outcome comparisons after the complete majority and unanimity objects have first been paired in the same primitive and belief fiber. Named proposals, named supports, and off-path strategy or belief functions are not invariant realized summaries and therefore remain in the underlying complete object. This proves the measurability, orbit-completeness, and universal anonymous factorization statements in Appendix F.1. $\square$

Appendix C: Endpoints, exact sets, and illustration

C.1 Endpoint equivalence

At $p = 0$, posterior support is the singleton low type. The private unanimity proposal and payoff coincide with the public low-type game. At $p = 1$, posterior support is the singleton high type, and the private pooling proposal and payoff coincide with the public high-type game. The same equivalence holds under majority: applying the relevant private class at each endpoint reproduces public inclusion or exclusion, including the $o = 1/m$ tie-break. Thus the endpoints are literal complete-information games, not one-sided limits.

C.2 Exact correspondence and envelopes

Most parameter cells are singleton payoff vectors. At $h = 1/m$, exact indifference between exclusion and pooling can leave a proposal segment. If λ is the exclusion weight, every reported object uses that same λ . The lower and upper envelope of any component may summarize the segment, but coordinated marginal envelopes do not imply that all pairwise combinations are attainable.

For majority-rule informational rents, the exact componentwise envelopes are

$$IR_M^B(\ell) \in [\min\{(1 - \beta)\ell, \beta(h - \ell)\}, \max\{(1 - \beta)\ell, \beta(h - \ell)\}], \quad IR_M^B(h) \in [0, (1 - \beta)h].$$

For both the private institutional payoff contrast and ΔIR^B , they are

$$\text{low type} \in [\min\{0, \beta h - \ell\}, \max\{0, \beta h - \ell\}], \quad \text{high type} \in [-(1 - \beta)h, 0].$$

In each case these are the marginal envelopes of a single line segment. The same λ indexes both coordinates, so the Cartesian product of the two intervals is not attainable.

Where $0 < p \leq p^*$, the private unanimity correspondence is empty. Accordingly, V_U^B , IR_U^B , and ΔIR^B are empty. No payoff, sign, institutional ranking, or interpolation is assigned to that cell.

Appendix D: Notation

Table 6: Notation for the essential-input bargaining game

Symbol	Meaning
H	Privately informed hegemon
W, m	Set and number of weak states; $m \geq 3$
$k = \lfloor (m + 1)/2 \rfloor$	Additional yes votes a proposer needs under majority
$o \in \{\ell, h\}$	Hegemon's terminal disagreement payoff, with $0 < \ell < h < 1$
p	Prior probability of the high outside option h
μ, μ^{off}	Posterior probability of h , on and off path
β	One-round discount factor, strictly between zero and one
$x = (x_H, x_1, \dots, x_m)$	Feasible allocation vector
x_H, x_j, x_i	Allocations to H , weak responder j , and proposer i
$w = \beta/m$	First-round continuation value of a weak state under majority
$w_o^U = \beta(1 - o)/m$	First-round continuation value of a weak state under unanimity
p^*	Terminal unanimity cutoff, $(h - \ell)/(1 - \ell)$
B, A	Games without and with the earlier hegemonic agenda stage
$v_g^B(o), v_g^A(o)$	Public-information payoff under rule g in each arm
$V_g^B(o), V_g^A(o)$	Private-information payoff correspondence in each arm
IR_g^B, IR_g^A	Private payoff minus the same-arm public benchmark
$\Delta IR^B, \Delta IR^A$	Unanimity-minus-majority informational-rent contrasts
$e = m - k$	Weak states excluded by a minimal majority coalition
v_M^{safe}	Safe majority-agenda payoff, $1 - k\beta/m$
o_M^*	Majority-agenda delay cutoff, v_M^{safe}/β
ρ	Off-path likelihood-ratio coordinate in the agenda extension
ea	Ex ante coordinate, evaluated as the prior-weighted average of the two types
$\Delta v^A(o)$	Public institutional gap with agenda, $v_U^A(o) - v_M^A(o)$
$\Delta V^A(o)$	Private institutional contrast with agenda, $V_U^A(o) - V_M^A(o)$
D_g, I_g, T_g	Direct public agenda effect, agenda–information interaction, and total private agenda effect

Symbol	Meaning
Q_g	Diagonal agenda-only versus information-only comparison

The symbols w and w_o^U price a weak vote inside the baseline arm. The functions $r_M(o)$ and $r_U(o)$ in Appendix E price a weak vote one date earlier, after rejection enters the baseline continuation. The extra discounting means that these are date-specific objects, not alternative names for the same quantity.

Appendix E: Exact agenda-stage correspondences and comparisons

E.1 Agenda-stage contract

Fix

$$\begin{aligned} \mathbf{d} &= (m, k, e, \beta, \ell, h, \mathcal{X}, p), \\ m &\geq 3, \quad k = \lfloor (m+1)/2 \rfloor, \quad e = m - k, \\ 0 &< \beta < 1, \quad 0 < \ell < h < 1, \quad p \in [0, 1]. \end{aligned}$$

At date A , after observing its type, H proposes

$$x^A = (x_H, x_1, \dots, x_m) \in \mathcal{X}, \quad \mathcal{X} = \{x^A : x_H \geq 0, x_j \geq 0, x_H + \sum_j x_j \leq 1\}.$$

The proposal counts as H 's affirmative vote. Weak states vote simultaneously and in pure strategies. Majority requires k affirmative weak votes; unanimity requires all m . Passage implements the package at date A . Rejection enters the corresponding frozen Round-1 continuation, whose native payoff is multiplied by β exactly once.

A rejected public history includes the rule, proposal, proposer, full ballot, outcome, and posterior. Its continuation selector is total, public, Borel, common to types compatible with the history, and returns one complete frozen Round-1 equilibrium record. A record binds strategies, beliefs, recognition lotteries, ballots, payoffs, and outcomes. No scalar continuation payoff may replace it.

For an interior prior, the equilibrium proposal laws of the two types are Borel measures σ_ℓ, σ_h . At every disciplined proposal, Bayes' rule determines the posterior from their local likelihood ratio. At undisciplined proposals, the M/S/B restriction fixes the posterior to

$$\mu^{\text{off}} = b_\rho(p) = \frac{p\rho}{1 - p + p\rho}, \quad \rho \in [0, \infty].$$

The extended-real boundary conventions are

$$b_0(p) = 0, \quad b_\infty(p) = 1$$

for an interior prior. The endpoints preserve prior support: $\mu^{\text{off}} = p$ everywhere, the fiber is $(*, p)$, and ρ is irrelevant. Voting follows the same as-if-pivotal comparison and indifference-to-yes convention as the baseline.

The single coordinate ρ is common to every undisciplined proposal. This is a stronger restriction than the ballot-by-ballot free values of Appendix A.2, adopted here to index the correspondences. The two conventions never interact: a rejected agenda proposal enters a complete Round-1 record, whose internal beliefs follow the baseline rule.

E.2 Complete private-majority correspondence

For each posterior μ and a fixed anonymous Markov continuation selection χ , let $a_\chi^{\text{pass}}(\mu)$ be H 's best payoff from a proposal that passes and let $d_{\chi,o}(\mu)$ be the payoff of a type with outside option o from rejection, both at date A . Define

$$a_\mu^{\text{pass}} := a_\chi^{\text{pass}}(\mu), \quad d_{o,\mu} := d_{\chi,o}(\mu), \quad \mu \in [0, 1],$$

so that $a_p^{\text{pass}}, a_0^{\text{pass}}, a_1^{\text{pass}}$ and $d_{\ell,p}, d_{h,p}, d_{\ell,0}, d_{\ell,1}, d_{h,0}, d_{h,1}$ denote evaluations of these same functions at the displayed posterior. Also define

$$O_o(\rho) = \max\{a_\chi^{\text{pass}}(b_\rho(p)), d_{\chi,o}(b_\rho(p))\}.$$

The following conditions are necessary and sufficient for every interior pure equilibrium class:

Table 7: Pure private-majority classes at the agenda stage

Class	Necessary and sufficient condition and linked type payoff
Pooling agreement	$O_h(\rho) \leq a_p^{\text{pass}}$, with any $z \in [O_h(\rho), a_p^{\text{pass}}]$; linked payoff (z, z)
Pooling delay	$d_{\ell,p} \geq O_\ell(\rho)$ and $d_{h,p} \geq O_h(\rho)$; linked payoff $(d_{\ell,p}, d_{h,p})$
Separating agreement–agreement	$O_h(\rho) \leq \min\{a_0^{\text{pass}}, a_1^{\text{pass}}\}$, with any $z \in [O_h(\rho), \min\{a_0^{\text{pass}}, a_1^{\text{pass}}\}]$; linked payoff (z, z)
Low agreement, high delay	$d_{h,1} \geq O_h(\rho)$ and $\max\{d_{\ell,1}, O_\ell(\rho)\} \leq \min\{a_0^{\text{pass}}, d_{h,1}\}$; linked payoff $(z, d_{h,1})$, $z \in [\max\{d_{\ell,1}, O_\ell(\rho)\}, \min\{a_0^{\text{pass}}, d_{h,1}\}]$
Low delay, high agreement	Impossible; linked payoff $\emptyset$
Separating delay–delay	$d_{\ell,0} \geq d_{\ell,1}$, $d_{h,1} \geq d_{h,0}$, $d_{\ell,0} \geq O_\ell(\rho)$, and $d_{h,1} \geq O_h(\rho)$; linked payoff $(d_{\ell,0}, d_{h,1})$

These classes preserve proposals and messages even when their payoffs coincide. Separating agreement at the same z is separation in messages at indifference, not pooling. The off-path feasibility set for ρ can be disconnected because the frozen majority continuation changes branches; no single cutoff substitutes for the table.

For mixed strategies, a complete reduced record is

$$R_M = (\rho, \mu^{\text{off}}, \sigma_\ell, \sigma_h, \bar{\sigma}, \pi, \chi, b_M, u_\ell, u_h).$$

Here $\bar{\sigma} = (1 - p)\sigma_\ell + p\sigma_h$ is the public proposal law. The record also retains posterior map π ,

continuation selection χ , ballot map b_M , and both type payoffs. Let

$$\begin{aligned} S_M &= \text{supp}(\bar{\sigma}), & u_o(x) &= b_M(x)x_H + \{1 - b_M(x)\}d_{\chi,o}(\pi(x)), \\ V_o &= \int u_o(x) \sigma_o(dx), & \mathfrak{D}_o(S_M) &= \sup_{x \notin S_M} u_o^{\text{off}}(x), \end{aligned}$$

where u_o^{off} evaluates an unsupported proposal at μ^{off} , and the supremum of the empty set is $-\infty$. In addition to pointwise Bayes updating on the disciplined support, the pivotal ballot rule at every proposal, and the continuation selection χ , the necessary-and-sufficient proposal conditions are

$$\begin{aligned} u_o(x) &\leq V_o && \text{for every } x \in S_M, \\ u_o(x) &= V_o && \text{for } \sigma_o\text{-almost every } x, \\ V_o &\geq \mathfrak{D}_o(S_M), && o \in \{\ell, h\}. \end{aligned}$$

Thus arbitrary Borel mixtures and atomless supports are included.

At $p \in \{0, 1\}$, the posterior is fixed at the endpoint. With

$$\bar{V}_o = \max\{a_p^{\text{pass}}, d_{o,p}\},$$

each type's Borel proposal law may be any probability measure supported on its argmax. The probability-zero type remains part of the linked record. For every admissible primitive and prior, at least one majority PBE exists for some ρ ; this does not assert existence at every fixed ρ . Appendix B.7 proves the pure enumeration, the mixed membership criterion, the endpoint correspondences, existence, and the uniform lower bound used in the body.

E.3 Complete private-unanimity correspondence

Define

$$r_U(o) = \frac{\beta(1 - \beta o)}{m}, \quad v_U^A(o) = 1 - \beta + \beta^2 o.$$

Thus $0 < r_U(h) < r_U(\ell)$, and the two proposal primitives are

$$x^\ell = (v_U^A(\ell), r_U(\ell), \dots, r_U(\ell)), \quad x^h = (v_U^A(h), r_U(h), \dots, r_U(h)).$$

At posterior zero, every weak state accepts exactly when $x_j \geq r_U(\ell)$; at a high posterior, it accepts exactly when $x_j \geq r_U(h)$. Equality passes under the indifference-to-yes convention. The only continuation posteriors admitted by the frozen unanimity continuation are

$$\mathcal{P}_C = \{0\} \cup (p^*, 1].$$

Rejection pays the high type $\beta^2 h$ at every admissible posterior; it pays the low type $\beta^2 \ell$ at posterior zero and $\beta^2 h$ at a high posterior.

A complete unanimity member is the indivisible binder

$$R_U = (\sigma_\ell, \sigma_h, \mu, \mu^{\text{off}}, \hat{\kappa}_U, b_U, \Omega, \Gamma_\ell, \Gamma_h).$$

Here σ_o is a Borel proposal law, $\bar{\sigma} = (1 - p)\sigma_\ell + p\sigma_h$, μ is the public posterior, and $\mu^{\text{off}} \in \mathcal{P}_C$ is its single value at every undisciplined proposal. The selector $\hat{\kappa}_U$ returns a complete literal frozen continuation member at the same posterior, b_U is the full pure ballot map, and $\Omega_o(x)$ is the literal terminal law induced by passage or by that continuation. Finally, Γ_o is the realized pushforward law of the joint record

$$(x, \mu(x), b_U(x), \text{pass/reject}, \hat{\kappa}_U, \Omega_o(x), \text{date-}A \text{ payoffs}).$$

The local Bayes limit must exist at every disciplined proposal, including zero-mass limit points, and must lie in $\mathcal{P}_C$. At each such point, and at every undisciplined point evaluated with μ^{off} , neither type may obtain a profitable proposal deviation. These pointwise restrictions, the ballot map, the literal continuation, and both realized laws are part of the same binder and cannot be recombined across members.

The full interior correspondence is generated by the following two families.

Low-posterior family AU-MSB-L. Its exact domain is

$$0 < p < 1, \quad v_U^A(\ell) \geq \beta^2 h, \quad \mu^{\text{off}} = 0.$$

Choose Borel laws satisfying all of the following:

1. $\sigma_\ell(\{x^\ell\}) = \alpha_L > 0$ and $\sigma_h(\{x^\ell\}) = 0$;
2. outside x^ℓ , $\mu(x) > p^*$ $\bar{\sigma}$ -almost surely;
3. each used proposal outside x^ℓ either passes with H 's share $v_U^A(\ell)$, or, only when $v_U^A(\ell) = \beta^2 h$, rejects and pays $\beta^2 h = v_U^A(\ell)$;
4. every disciplined proposal, including every zero-mass limit point, gives each type a deviation payoff no larger than $v_U^A(\ell)$; every undisciplined proposal satisfies the same inequality under posterior zero.

Every accepted high-posterior signal therefore belongs to

$$\mathcal{K}_L = \{x \in \mathcal{X} : x_H = v_U^A(\ell), \min_j x_j \geq r_U(h)\}.$$

The laws may be asymmetric, discrete, semi-pooling, mixed, or atomless. A pure witness has the low type propose x^ℓ and the high type propose $x^S = (v_U^A(\ell), r_U(h), \dots, r_U(h))$. Every member has linked payoff $(v_U^A(\ell), v_U^A(\ell))$. If $v_U^A(\ell) > \beta^2 h$, agreement occurs with probability one for both types; if $v_U^A(\ell) = \beta^2 h$, high-posterior rejected signals may carry mass. The weak-state payoff attached to the same member is x_j after passage, $r_U(\ell)$ after rejection at $(\mu, o) = (0, \ell)$, zero after rejection at $(0, h)$, and $r_U(h)$ after rejection at any high posterior.

High-posterior family AU-MSB-H. Its necessary prior domain is $p^* < p < 1$. When $\mu^{\text{off}} = 0$, choose any

$$u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)],$$

and Borel laws such that: $\mu > p^*$ $\bar{\sigma}$ -almost surely; the local admissible Bayes limit exists at every disciplined point; every used accepted proposal lies in

$$\mathcal{K}_H(u) = \{x \in \mathcal{X} : x_H = u, \min_j x_j \geq r_U(h)\};$$

used rejected proposals occur only when $u = \beta^2 h$; every disciplined deviation gives each type at most u ; and every undisciplined deviation, evaluated at posterior zero, also gives at most u . The linked payoff is (u, u) . If $u > \beta^2 h$, agreement has probability one for each type; if $u = \beta^2 h$, each type may mix accepted proposals with $x_H = \beta^2 h$ and rejected high-posterior proposals. A pure

pooling witness for every u is

$$x(u) = \left(u, \frac{1-u}{m}, \dots, \frac{1-u}{m} \right), \quad \sigma_\ell = \sigma_h = \delta_{x(u)}, \quad \mu(x(u)) = p.$$

When $\mu^{\text{off}} \in (p^*, 1]$, the only current proposal laws are $\sigma_\ell = \sigma_h = \delta_{x^h}$, with $\mu(x^h) = p$ and payoff $(v_U^A(h), v_U^A(h))$; multiplicity survives only in μ^{off} and the complete literal continuation binder. When $\mu^{\text{off}} \in (0, p^*]$, the correspondence is empty. For $p^* < p < 1$ and $\mu^{\text{off}} = 0$, the high-posterior family always exists, and the low-posterior family is additionally present exactly when $v_U^A(\ell) \geq \beta^2 h$. For $0 < p \leq p^*$, equilibrium exists exactly when $v_U^A(\ell) \geq \beta^2 h$ and consists only of the low-posterior family. There is no third interior family.

The endpoints are complete Borel strategy correspondences, not one-sided limits. At $p = 0$, support preservation fixes $\mu(x) = \mu^{\text{off}} = 0$ everywhere, $\sigma_\ell = \delta_{x^\ell}$, and the counterfactual high-type law is

$$\begin{cases} \delta_{x^\ell}, & v_U^A(\ell) > \beta^2 h, \\ \text{any Borel law supported on } \{x^\ell\} \cup \mathcal{R}_L, & v_U^A(\ell) = \beta^2 h, \\ \text{any Borel law supported on } \mathcal{R}_L, & v_U^A(\ell) < \beta^2 h, \end{cases}$$

where

$$\mathcal{R}_L = \{x \in \mathcal{X} : \min_j x_j < r_U(\ell)\}.$$

The linked payoff is $(v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\})$, and the low type agrees immediately with probability one. At $p = 1$, support preservation fixes $\mu(x) = \mu^{\text{off}} = 1$ everywhere and $\sigma_\ell = \sigma_h = \delta_{x^h}$, including the probability-zero type's strategy, with linked payoff

$$V_U^A = (v_U^A(h), v_U^A(h)).$$

Consequently, the payoff image of the complete correspondence is

$$V_U^A = \begin{cases} (v_U^A(\ell), \max\{v_U^A(\ell), \beta^2 h\}), & p = 0, \\ (v_U^A(\ell), v_U^A(\ell)), & 0 < p \leq p^*, v_U^A(\ell) \geq \beta^2 h, \mu^{\text{off}} = 0, \\ \{(u, u) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1, \mu^{\text{off}} = 0, \\ (v_U^A(h), v_U^A(h)), & p^* < p < 1, \mu^{\text{off}} \in (p^*, 1], \\ (v_U^A(h), v_U^A(h)), & p = 1, \\ \emptyset, & \text{all remaining fibers.} \end{cases}$$

This display is only an image: it neither replaces the binder nor converts an interval of linked vectors into a Cartesian product.

For the two-layer representation, let Z_U be the compact realized-record space containing the named proposal and ballot coordinates, the realized posterior, pass/reject, the literal continuation-cell identity, and the terminal allocation and payoff record. Then

$$\Gamma_o^{U, R_U} = \text{Law}_o(x, \mu(x), b_U(x), \text{pass/reject}, \zeta_U, \omega_{\text{term}}) \in \mathcal{P}(Z_U).$$

Under a common permutation τ of weak-state names, let $\mathcal{R}_\tau$ relabel every named weak-state coordinate and let Λ_γ code the diagonal orbit of $\gamma = (\Gamma_\ell, \Gamma_h)$. For interior priors,

$$\text{Sig}_U^{\text{ex}}(R_U) = (\rho, \mu^{\text{off}}, \Lambda_\gamma), \quad \text{Sum}_U^{\text{econ}}(R_U) = (\rho, \mu^{\text{off}}, (q_U)_\# \Gamma_\ell, (q_U)_\# \Gamma_h),$$

with endpoint forms $(*, p, \Lambda_\gamma)$ and $(*, p, (q_U)_\# \Gamma_\ell, (q_U)_\# \Gamma_h)$. The exact signature preserves the diagonal orbit of realized laws and their relevant realized identities; the economic layer removes names record by record. Neither layer contains the full off-path functions. Formal-identity and realized-law operations use the exact signature; any operation that depends on off-path beliefs, strategies, supports, messages, or continuation plans must instead consume the underlying complete binder. No Reynolds average is treated as an equilibrium representative, and every none cell above remains empty in both layers. Appendix B.8 proves the two-family characterization, its exhaustiveness and existence domains, the endpoint correspondences, and the payoff bound used in

the body.

E.4 Exact same-fiber institutional comparison

Let $\mathcal{B}_M(\mathbf{d}, \rho, \mu^{\text{off}})$ and $\mathcal{B}_U(\mathbf{d}, \rho, \mu^{\text{off}})$ be the full correspondences just described. The admissible comparison set is

$$\mathcal{J}_A^{\text{bind}}(\mathbf{d}, \rho, \mu^{\text{off}}) = \mathcal{B}_M(\mathbf{d}, \rho, \mu^{\text{off}}) \times_{(\mathbf{d}, \rho, \mu^{\text{off}})} \mathcal{B}_U(\mathbf{d}, \rho, \mu^{\text{off}}).$$

This product requires identical primitives and the same fiber, but it imposes no common proposal, continuation realization, or cross-world randomization. The exact linked type-payoff contrast is

$$\mathcal{C}_{\ell h}^A(\mathbf{d}, \rho, \mu^{\text{off}}) = \{u - v : u \in V_U^A(\mathbf{d}, \rho, \mu^{\text{off}}), v \in V_M^A(\mathbf{d}, \rho, \mu^{\text{off}})\}.$$

It is a difference of complete vectors. The ex ante image is

$$\mathcal{C}_{\text{ea}}^A(\mathbf{d}, \rho, \mu^{\text{off}}) = \{(1 - p)x_\ell + px_h : (x_\ell, x_h) \in \mathcal{C}_{\ell h}^A(\mathbf{d}, \rho, \mu^{\text{off}})\}.$$

It is generally not equal to the sum of independently chosen marginal images. Dominance claims are setwise: unanimity dominates in a coordinate only when every member has positive sign; majority dominates only when every member has negative sign. A crossing set is selection dependent. An empty fiber has classification none, not sign zero.

E.5 The sufficient majority-advantage region

Every majority binder satisfies

$$V_M^A(o) \geq \max\{v_M^{\text{safe}}, \beta^2 o\}, \quad v_M^{\text{safe}} = 1 - \frac{k\beta}{m},$$

while every unanimity binder satisfies

$$V_U^A(o) \leq v_U^A(h) = 1 - \beta + \beta^2 h.$$

Therefore

$$v_M^{\text{safe}} - v_U^A(h) = \beta \left(\frac{e}{m} - \beta h \right).$$

If $\beta h < e/m$, then this margin is positive and every comparable member satisfies

$$V_U^A(o) - V_M^A(o) \leq -\beta \left(\frac{e}{m} - \beta h \right) < 0$$

for both types and ex ante. Equality yields weak majority advantage. This proves sufficiency only.

The condition is not necessary. The following separate parameter point is chosen to demonstrate that failure of necessity, rather than to replace the running illustration. With

$$m = 4, k = 2, e = 2, \beta = .9, \ell = .5, h = .6, p = 0,$$

we have $e/m = .5$ and $\beta h = .54 > .5$, so the condition fails. Nevertheless $v_M^{\text{safe}} = .55$, $v_U^A(\ell) = .505$, and $\beta^2 h = .486$; hence every majority binder strictly exceeds the endpoint unanimity vector $(.505, .505)$.

The excluded fraction is

$$\frac{e}{m} = \begin{cases} 1/2, & m \text{ even,} \\ (m-1)/(2m), & m \text{ odd.} \end{cases}$$

In an existing low-prior cell, the weaker local condition $\beta \ell < e/m$ gives the margin $\beta(e/m - \beta \ell)$. At $p = 0$, that condition guarantees only the ex ante conclusion unless the full counterfactual type vector is used.

E.6 Public-majority PBE classes

For public $o \leq 1/m$, each weak vote costs

$$r_M(o) = \frac{\beta(1 - \beta o)}{m}.$$

Every optimal proposal pays exactly $r_M(o)$ to exactly k weak states, pays zero to the rest, and leaves

$$v_M^A(o) = 1 - \frac{k\beta(1 - \beta o)}{m}$$

to H . Every probability measure over these minimal coalitions is a PBE proposal lottery. Passage strictly dominates the delayed payoff $\beta^2 o$.

For $o > 1/m$, every purchased weak vote costs β/m , so passage leaves v_M^{safe} and rejection yields βo . Thus:

- $1/m < o < o_M^*$: every PBE passes through a minimal coalition,
- $o = o_M^*$: every linked mixture of minimal passage and delay is allowed,
- $o > o_M^*$: every PBE deliberately delays.

At the tie, the same mixture weight binds the proposal, result, and agreement probability. The public payoff is always the singleton

$$v_M^A(o) = \begin{cases} 1 - \frac{k\beta(1 - \beta o)}{m}, & o \leq 1/m, \\ \max\{v_M^{\text{safe}}, \beta o\}, & o > 1/m. \end{cases}$$

E.7 Public-unanimity PBE

Every weak vote costs

$$r_U(o) = \frac{\beta(1 - \beta o)}{m}.$$

The unique optimal on-path proposal pays this amount to all m weak states and leaves

$$v_U^A(o) = 1 - \beta + \beta^2 o$$

to H . Rejection yields $\beta^2 o$, so passage is strictly better by $1 - \beta$. Agreement is immediate and payoff-determined for every public type; no majority-style delay is imported into this rule.

E.8 The public gap

Subtracting E.6 from E.7 gives the unanimity-minus-majority gap

$$\Delta v^A(o) = \begin{cases} -\beta(e/m)(1 - \beta o), & o \leq 1/m, \\ \beta(\beta o - e/m), & o > 1/m, \beta o \leq v_M^{\text{safe}}, \\ (1 - \beta)(1 - \beta o), & o > 1/m, \beta o \geq v_M^{\text{safe}}. \end{cases}$$

Continuity holds at $\beta o = v_M^{\text{safe}}$. For $o > 1/m$,

$$\text{sgn } \Delta v^A(o) = \text{sgn}(\beta o - e/m).$$

At $o = 1/m$, the first branch applies because the frozen no-agenda continuation includes H at equality. This special selection also matters for the direct agenda effect in E.12.

E.9 Exact informational-rent translations

For every complete private binder,

$$IR_g^A = V_g^A - (v_g^A(\ell), v_g^A(h)).$$

This fixed-vector translation preserves every atom, mixture, and linked type coordinate. The exact unanimity cells are:

Table 8: Unanimity informational rent with hegemonic agenda

Fiber	IR_U^A	Ex ante image
$p = 0$	$(0, \max\{v_U^A(\ell), \beta^2 h\} - v_U^A(h))$	0
$0 < p \leq p^*, v_U^A(\ell) \geq \beta^2 h, \mu^{\text{off}} = 0$	$(0, -(v_U^A(h) - v_U^A(\ell)))$	$-p(v_U^A(h) - v_U^A(\ell))$
Other low-prior fibers	$\emptyset$	$\emptyset$
$p^* < p < 1, \mu^{\text{off}} = 0$	$\{(u - v_U^A(\ell), u - v_U^A(h)) : u \in [\max\{v_U^A(\ell), \beta^2 h\} - v_U^A(\text{ea}(p)), v_U^A(h) - v_U^A(\text{ea}(p))]\}$	$(1 - p)(v_U^A(h) - v_U^A(\ell))$
$p^* < p < 1, \mu^{\text{off}} \in (p^*, 1]$	$(v_U^A(h) - v_U^A(\ell), 0)$	$v_U^A(\ell)$
$p^* < p < 1, \mu^{\text{off}} \in (0, p^*]$	$\emptyset$	$\emptyset$

Fiber	IR_U^A	Ex ante image
$p = 1$	$(v_U^A(h) - v_U^A(\ell), 0)$	0

Here $v_{U,ea}^A(p) = (1 - p)v_U^A(\ell) + pv_U^A(h)$. Every existing cell satisfies

$$IR_U^A(\ell) \geq 0, \quad IR_U^A(h) \leq 0,$$

with at least one strict inequality. Majority rent is retained as the exact translation of the full majority payoff correspondence; it has no imposed global sign.

E.10 Public power and information decomposition

For every same-fiber member,

$$\Delta IR^A(o) = IR_U^A(o) - IR_M^A(o) = \Delta V^A(o) - \Delta v^A(o),$$

or equivalently

$$\Delta V^A(o) = \Delta v^A(o) + \Delta IR^A(o).$$

If $\Delta v^A(\ell) < 0$ and a private member has $\Delta V^A(\ell) > 0$, then

$$\Delta IR^A(\ell) > -\Delta v^A(\ell) > 0$$

for that member. Type incidence precedes ex ante aggregation:

$$\Delta V_{ea}^A = \Delta v_{ea}^A + \Delta IR_{ea}^A, \quad \Delta v_{ea}^A = (1 - p)\Delta v^A(\ell) + p\Delta v^A(h).$$

Under the sufficient condition in E.5,

$$\Delta V^A(o) \leq -\beta(e/m - \beta h), \quad \Delta IR^A(o) \leq -\Delta v^A(o) - \beta(e/m - \beta h).$$

This bounds the informational difference relative to the amount required to reverse majority's guaranteed private advantage; it does not assign a sign to $\Delta IR^A(o)$.

The numerical member in Figure 5 gives

$$\Delta IR^A(\ell) = .5480, \quad \Delta v^A(\ell) = -.4095, \quad \Delta V^A(\ell) = .1385.$$

These linked values illustrate the identity; they do not calibrate the WTO or select globally from either correspondence.

E.11 Factorial design and identity

At date A , the four cells are

	B	A
public type	$\beta v_g^B(o)$	$v_g^A(o)$
private type	$\beta V_g^B(o)$	$V_g^A(o)$.

Define

$$\begin{aligned} D_g(o) &= v_g^A(o) - \beta v_g^B(o), \\ T_g(o) &= V_g^A(o) - \beta V_g^B(o), \\ I_g(o) &= IR_g^A(o) - \beta IR_g^B(o). \end{aligned}$$

Substituting $V = v + IR$ in both arms yields, by type and ex ante,

$$T_g = D_g + I_g.$$

The factor β appears once because the no-agenda arm begins one date later. The treatment combines an earlier opportunity with a mandatory proposal. Redating Round 1 to date A in the control would define another structural contrast.

E.12 Direct agenda effects and their institutional difference

The rule-specific public effects are

$$D_U(o) = 1 - \beta$$

and

$$D_M(o) = \begin{cases} v_M^{\text{safe}} - (e/m)\beta^2 o, & o \leq 1/m, \\ v_M^{\text{safe}} - \beta o, & 1/m < o < o_M^*, \\ 0, & o \geq o_M^*. \end{cases}$$

The threshold satisfies $o_M^* > 1/m$, but it can be weakly above one, in which case the last branch is empty in the admissible domain. At $o = 1/m$, the first branch applies. The frozen continuation selection creates the jump

$$D_M(1/m) - \lim_{o \downarrow 1/m, o > 1/m} D_M(o) = \frac{\beta}{m} \left(1 - \frac{e\beta}{m}\right) > 0.$$

With $\Delta D(o) = D_U(o) - D_M(o)$,

$$\Delta D(o) = \begin{cases} -\beta(e/m)(1 - \beta o), & o \leq 1/m, \\ \beta(o - e/m), & 1/m < o \leq o_M^*, \\ 1 - \beta, & o \geq o_M^*. \end{cases}$$

The last two formulas coincide at o_M^* . This branchwise statement is the frozen result; no stronger global classification from the later external consultation is promoted here.

E.13 Total private agenda effects

Under unanimity, the exact linked correspondence is

$$T_U = \begin{cases} (1 - \beta, \max\{v_U^A(\ell) - \beta^2 h, 0\}), & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ \{(u - \beta^2 h, u - \beta^2 h) : u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)]\}, & p^* < p < 1, \mu^{\text{off}} = 0, \\ \emptyset, & p^* < p < 1, \mu^{\text{off}} \in (0, p^*], \\ (1 - \beta, 1 - \beta), & p^* < p < 1, \mu^{\text{off}} \in (p^*, 1], \\ (1 - \beta, 1 - \beta), & p = 1. \end{cases}$$

In the low-prior interior, the no-agenda control is always empty; the agenda treatment may exist or may also be empty. In the high-prior low-positive off-path fiber, the control exists but the agenda treatment is empty. These different missing arms both propagate to an empty T_U .

Where both arms exist, every unanimity member weakly benefits both types and the maximum gain is $1 - \beta$. At $p = 0$, the ex ante gain is $1 - \beta$ even if the probability-zero high coordinate is zero.

For majority, retain the exact interaction correspondence:

$$T_M = D_M + I_M.$$

For each coordinate and member,

$$\begin{aligned} T_M(o) > 0 &\iff I_M(o) > -D_M(o), \\ T_M(o) = 0 &\iff I_M(o) = -D_M(o), \\ T_M(o) < 0 &\iff I_M(o) < -D_M(o). \end{aligned}$$

No selection-free sign follows when the source correspondence crosses the threshold. Across rules,

$$\Delta T(o) = \Delta D(o) + \Delta I(o)$$

member by member in every complete product. A robust ranking requires the entire admissible correspondence to lie on one side of $-\Delta D(o)$.

E.14 The diagonal contrast

The diagonal comparison is

$$Q_g(o) = v_g^A(o) - \beta V_g^B(o) = D_g(o) - \beta IR_g^B(o).$$

It compares public information with agenda against private information without agenda, changing two factors at once. It is not a causal effect of agenda or information alone.

Under unanimity,

$$Q_U = \begin{cases} (1 - \beta, 1 - \beta), & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ (v_U^A(\ell) - \beta^2 h, 1 - \beta), & p^* < p \leq 1. \end{cases}$$

The ex ante value in the high-prior region is

$$Q_{U,ea} = 1 - \beta - (1 - p)\beta^2(h - \ell).$$

Under majority,

$$Q_M = D_M - \beta IR_M^B$$

is the exact image of the applicable linked no-agenda records and remains set-valued where those records do. Q_U may exist where T_U is empty because it does not use the private agenda arm.

Appendix F: Factorization, interaction, and scope

F.1 Exact and economic layers

For majority, the exact signature is

$$\text{Sig}_M^{ex}(R_M) = (\rho, \mu^{\text{off}}, \Lambda_{(\Gamma_\ell, \Gamma_h)}),$$

where Λ codes the diagonal relabeling orbit of the two complete realized laws. Its economic summary is

$$\text{Sum}_M^{econ}(R_M) = (\rho, \mu^{\text{off}}, (q_Z)_\# \Gamma_\ell, (q_Z)_\# \Gamma_h).$$

Unanimity uses the analogous pair Sig_U^{ex} and Sum_U^{econ} . The exact signature preserves the diagonal relabeling orbit of the realized laws and the relevant identities inside those realized records. It does not retain the complete off-path belief, strategy, support, message, or continuation functions. Those functions remain only in the underlying complete binder, and every off-path-sensitive operation must consume that binder. The economic layer retains the anonymous realized laws needed for payoffs, agreement, delay, and anonymous outcomes.

The institutional operator is first applied to complete binders. Only after that product is formed do payoff and outcome maps factor measurably through the economic summaries:

$$\mathcal{C}_{\text{econ}}(R_M, R_U) = \bar{\mathcal{C}}_{\text{econ}}(\text{Sum}_M^{\text{econ}}(R_M), \text{Sum}_U^{\text{econ}}(R_U)).$$

No Reynolds average is an equilibrium representative. Equality of summaries does not identify named proposals, coalition lotteries, or off-path plans. Appendix B.9 proves that the exact orbit law is a complete invariant and that every anonymous Borel outcome used here factors uniquely through the economic summary.

For coordinate $r \in \{\ell, h, \text{ea}\}$, the contrast envelope satisfies

$$\inf \mathcal{C}_r^A = \inf U_r - \sup M_r, \quad \sup \mathcal{C}_r^A = \sup U_r - \inf M_r.$$

The interval between these numbers is an interval hull, not necessarily an attained set. Likewise, the outcome object is the ordered pair of marginal laws

$$\mathcal{O}_A(\mathbf{d}, \rho, \mu^{\text{off}}) = \{((\bar{\Gamma}_M^\ell, \bar{\Gamma}_M^h), (\bar{\Gamma}_U^\ell, \bar{\Gamma}_U^h)) : (R_M, R_U) \in \mathcal{J}_A^{\text{bind}}(\mathbf{d}, \rho, \mu^{\text{off}})\}.$$

It is not a joint counterfactual distribution. A common cross-world draw or coordinate splicing would add a primitive absent from the model.

F.2 Interaction with the no-agenda benchmark

Let IR_g^B be the native Round-1 rent vector from the no-agenda benchmark. The exact interaction is the Minkowski difference of complete linked records,

$$I_g = IR_g^A - \beta IR_g^B.$$

For majority, the applicable no-agenda records are the three public regions used in Proposition 5.6; no general sign is inferred.

Under unanimity:

$$I_U = \begin{cases} (0, \max\{v_U^A(\ell), \beta^2 h\} - v_U^A(h)), & p = 0, \\ \emptyset, & 0 < p \leq p^*, \\ \left\{ (u - v_U^A(h), u - v_U^A(h)) : \right. \\ \quad \left. u \in [\max\{v_U^A(\ell), \beta^2 h\}, v_U^A(h)] \right\}, & p^* < p < 1, \mu^{\text{off}} = 0, \\ (0, 0), & \begin{matrix} p^* < p < 1, \\ \mu^{\text{off}} \in (p^*, 1], \end{matrix} \\ \emptyset, & \begin{matrix} p^* < p < 1, \\ \mu^{\text{off}} \in (0, p^*], \end{matrix} \\ (0, 0), & p = 1. \end{cases}$$

Thus agenda weakly reduces unanimity's informational rent in every existing high-prior fiber; the change is strict for both types exactly when $\mu^{\text{off}} = 0$ and $u < v_U^A(h)$. At $p = 0$, only the probability-zero high coordinate falls.

The institutional interaction is

$$\Delta I = \Delta IR^A - \beta \Delta IR^B.$$

It exists only when both exact sources exist. It remains set-valued whenever either source does and is never assembled from independently chosen marginal coordinates.

F.3 Payoff dates, existence, and verification limits

All agenda-game values arrive at date A . Native no-agenda Round-1 values are multiplied by β exactly once before comparison. Public agenda payoffs, private agenda payoffs, agenda rents, and their institutional contrasts receive no additional factor.

The existence rules are:

- every public agenda game has a PBE for both rules and types;
- IR_g^A exists exactly where the private agenda source exists;
- ΔIR^A exists exactly where the same-fiber product exists;
- I_g exists exactly where both its agenda and no-agenda rent sources exist;

- ΔI , T_g , and ΔT exist only where every source required by their definitions exists.

An absent source propagates $\emptyset$. It is never encoded as zero, NA, infinity, or a fictional payoff. Appendices B.7–B.9 prove the completeness and existence claims for the agenda correspondences, the absence of a third unanimity family, and the measurability and factorization properties used in the comparison. No conventional value is assigned to an empty source.

F.4 Causal scope and mandatory treatment

T_g is a structural causal contrast inside the specified model. It is not an empirically identified estimand for the WTO or another institution. The treatment inserts an earlier and mandatory hegemonic proposal stage. Its control starts at Round 1, so T_g includes the value of timing as specified by the game. It does not describe an optional proposal right that H can decline while retaining the same date- A payoff.

When a contrast is set-valued, each comparison is member-specific. A selection-free claim requires the entire exact correspondence to lie on the relevant side of the threshold. We do not promote the six later advisory corollaries to frozen theorems. The exact result remains the source correspondence and its declared domain.

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